

Improving Trust in Credit Scoring Using Explainable Artificial Intelligence

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ABSTRACT

This study looks at whether SHAP and LIME, the two most common tools for explaining machine learning predictions, actually give stable and trustworthy explanations when used for credit scoring. Four models were trained on the Home Credit Default Risk dataset: logistic regression, a decision tree, random forest, and XGBoost. Each model's predictions were explained using SHAP and LIME, and the resulting explanations were tested two ways: first by making small changes to an applicant's data to see if the explanation held up, and second by comparing explanations given to genuinely similar applicants. Logistic regression produced the most stable explanations under both methods and both tests, while the decision tree's SHAP explanations were the least consistent. LIME was more consistent than SHAP across every model tested. The study also found no trade-off between predictive performance and explanation stability: the best-performing model, logistic regression, was also the most stable. These results show that explanation stability depends heavily on which model and which explanation method are used, and that this matters as much as raw accuracy in a domain like credit scoring, where explanations may need to be defended to a regulator or an applicant.

Key words:

Credit Scoring, Explainable AI, SHAP, LIME, Explanation Stability, Logistic Regression, XGBoost, Home Credit Default Risk

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Chapter 1: Introduction

1.1 Motivation

Credit scoring is widely used by financial institutions to assess whether a borrower is likely to repay a loan. Traditionally, statistical models such as logistic regression have been used due to their simplicity and interpretability. However, with the growth of complex financial data, machine learning models such as Random Forest, Gradient Boosting, and LightGBM, are increasingly used because they provide a higher predictive accuracy.

Despite their performance, these models are often considered black-box systems, meaning their decision-making process is not easily understood. This lack of transparency creates some challenges in financial applications, where the decisions must be explainable to the regulators and customers. Explainable Artificial Intelligence (XAI) has emerged as a solution by providing techniques such as SHAP and LIME to interpret the model's predictions. However, while these methods improve transparency, their reliability and consistency are not always evaluated.

1.2 Problem Statement

Although explainable AI methods are widely used in credit scoring, most studies focus on prediction accuracy or basic interpretability. There is limited research on whether these explanations are stable and consistent. Explanations may vary even when the input data changes slightly, which can reduce the confidence in the model. In high-stakes domains such as finance, inconsistent explanations can undermine the trust in automated decisions. It is therefore necessary to evaluate whether XAI methods produce reliable and consistent explanations in credit scoring systems.

1.3 Objectives

The objectives of this study are to improve the trust in credit scoring systems by evaluating the stability and consistency of XAI methods. The concerned tasks are as follows:

- Develop credit scoring models using interpretable and machine learning techniques.
- Apply SHAP and LIME to explain the model predictions.
- Evaluate the stability of explanations across similar inputs.
- Compare the explanation consistency across different models and methods.
- Analyze the relationship between the model performance and the explanation reliability

1.4 Research Questions

The following research questions guide this study:

- How consistent are the SHAP and LIME explanations for the same applicant profile under small input perturbations?
- Do similar applicants receive similar explanations under the same trained credit-scoring model?
- How does the explanation stability differ between the Logistic Regression and black-box models such as XGBoost?
- Does a higher predictive performance correspond to more stable explanations? or is there a trade-off between the predictive accuracy and the explanation stability?

1.5 Considered Dataset

This study uses the Home Credit Default Risk dataset from Kaggle, which is a real-world credit dataset that consists of a main application table together with several supporting tables describing the applicants' external credit bureau history, previous applications with the same lender, and prior instalment repayment behaviour. Chapter 3 describes in detail how these tables were combined and prepared, and what the final resulting dataset shows.

1.6 Expected Outcomes

The outputs of this report are as follows:

- Development of accurate credit scoring models.
- Application of XAI techniques for interpretation.
- Evaluation of the stability and consistency of explanations.
- Identification of limitations in the current XAI methods.
- Recommendations for improving the reliability in explainable credit scoring systems.

1.7 Report Organization

This report is organized as follows:

- Chapter 1 introduces the motivation, problem definition, objectives, research questions and considered dataset, and expected outcomes.
- Chapter 2 is devoted to the literature review. Representative research studies on XAI in finance are described, along with the addressed research gap.
- Chapter 3 describes the design of the considered dataset, along with its preprocessing and an exploratory data analysis.
- Chapter 4 describes the considered methodology, that consists of preparing the features set from the considered dataset, developing the credit scoring models, generating SHAP and LIME explanations for these models, and designing the stability tests that answer the considered research questions.
- Chapter 5 presents the findings from our methodology, that consists of a predictive performance of the trained models, two stability tests, and a statistical comparison of stability across the models.
- Chapter 6 concludes the report and highlight some potential future works pertaining to its scope.

Chapter 2: Literature Review

Credit scoring sits at the intersection of two competing pressures in modern machine learning: the drive toward higher predictive accuracy, typically achieved through complex ensemble and black-box models, and the need for transparency in the decisions that materially affect the people’s financial lives. This review situates the present study within the growing body of research on XAI in finance, with particular attention to the methods that are most relevant to this project: SHAP and LIME, and to a body of work that this study argues has been comparatively under-examined: whether the explanations that these methods produce are themselves stable and trustworthy, rather than simply available.

This chapter is organized as follows. Section 2.1 sets out the case for explainable AI in financial services, covering the regulatory and practical pressures driving its adoption. Section 2.2 reviews explainable AI methods as they have been applied specifically to credit scoring. Section 2.3 introduces the two core explainability techniques central to this study, SHAP and LIME. Section 2.4 examines the stability and reliability gap in post-hoc explanations identified in the wider machine learning literature. Section 2.5 discusses the progression from interpretable baseline models to black-box ensembles that motivates the model comparison in this study. Section 2.6 synthesises these strands of literature and highlights the research gap this study addresses.

2.1 The Case for Explainable AI in Financial Services

The push toward explainability in finance is driven by both the regulatory and practical concerns. Goodman and Flaxman (2017) argue that the European Union’s General Data Protection Regulation effectively establishes a right to explanation for individuals subject to automated decisions that significantly affect them, placing a pressure on financial institutions to justify the algorithmic credit decisions rather than treating them as unreviewable outputs of a black box. This regulatory dimension is echoed throughout the finance-specific XAI literature, which frames explainability not as an optional enhancement, but as a precondition for the adoption of advanced machine learning in regulated lending environments.

Černevičienė and Kabašinskas (2024) provide the broadest evidence for this trend, systematically reviewing 138 articles on XAI in finance published between 2005 and 2022. Their review identifies credit management as one of the most common financial applications of XAI, alongside the stock price prediction and fraud detection, and concludes that explainability is essential for building trust in automated decisions, for detecting algorithmic bias, and supporting the regulatory oversight. Notably, their analysis finds that simpler and inherently interpretable methods remain more common in financial practices than complex multi-layer models, even where the latter may offer a superior raw predictive performance, a finding that itself motivates a closer scrutiny of whether post-hoc explanation methods can responsibly close that gap without introducing new, less visible problems of their own.

2.2 Explainable AI Methods Applied to Credit Scoring

A substantial and growing body of applied work has paired black-box credit-scoring models with post-hoc explanation methods, typically SHAP or LIME, to reconcile the predictive performance with interpretability. De Lange et al. (2022) combined a LightGBM model with SHAP to predict the default on a dataset of unsecured consumer loans from a Norwegian bank, finding that the resulting model outperformed the bank’s existing logistic regression scorecard while still surfacing the explanatory variables driving the individual predictions. Nallakaruppan et al. (2024) similarly proposed an explainable AI framework for credit evaluation that integrates the feature attribution directly into the credit decision pipeline, arguing that predictive accuracy and interpretability are too often treated as separate objectives rather than jointly optimizing the requirements of a deployable system.

Bussmann et al. (2021) extended this line of work in a direction especially relevant to the present study: rather than treating each SHAP explanation in isolation, they apply the correlation networks to the Shapley values across a sample of over 15,000 small and medium-sized enterprises, grouping the borrowers according to the similarity of their underlying explanations rather than their raw features alone. Their approach implicitly assumed that similar applicants should indeed receive similar explanations, the same assumption this study sets out to test directly rather than presuppose. Gramegna and Giudici (2021) took a more explicitly comparative approach, which consisted of evaluating the discriminative power of SHAP and LIME against one another in a credit risk setting and finding meaningful differences in how the two methods rank the features importance, a result that foreshadows the stability questions explored later in this review.

2.3 Core Explainability Techniques: SHAP and LIME

The two explanation methods central to this study both originate from work seeking a principled, model-agnostic way to attribute a prediction to its input features. Ribeiro et al. (2016) introduced Local Interpretable Model-agnostic Explanations (LIME), which approximates the behaviour of any classifier in the local neighbourhood of a specific prediction using a simpler, interpretable surrogate model fitted to the perturbed samples around that instance. Lundberg and Lee (2017) subsequently proposed SHapley Additive exPlanations (SHAP), which unifies LIME and several attribution methods within a single game-theoretic framework grounded in Shapley values from cooperative game theory. They demonstrated that this framework yields a unique solution satisfying a specific set of desirable theoretical properties that many prior methods lack.

Both methods have become close to the default choice for explaining black-box models in high-stakes domains, including credit scoring because they do not require access to a model’s internal structure and they can, in principle, be applied to any classifier. This model-agnostic flexibility is a major part of their appeal, but it is also the property that the later work identifies as a source of fragility. Indeed, both methods rely on sampling or perturbing the input space around an instance, and the resulting explanation depends on choices such as sampling procedure, kernel width, and background distribution, that are somewhat arbitrary and can therefore vary between runs or under small changes to the input in ways that a model’s actual prediction does not.

2.4 Stability and Reliability Gap in Post-Hoc Explanations

While the aforementioned finance-specific literature reviewed above largely treats SHAP and LIME as reliable once applied, a separate and more general strand of the explainability literature raises some doubts about that reliability, which have been explored far more often on images and text data than on structured, tabular financial data of the kind used in credit scoring. Alvarez-Melis and Jaakkola (2018) argued that robustness - the property that similar inputs should produce similar explanations - is a core requirement for any interpretability method to be considered trustworthy, and they introduced some quantitative metrics to measure it. Applying these metrics to several widely used interpretability approaches, they found that none perform well by this standard, and proposed that robustness should be actively enforced rather than assumed.

Zhang et al. (2019) focused specifically on LIME and identified multiple independent sources of uncertainty in its explanations: the randomness inherent in its sampling procedure, and the systematic variation in explanation quality depending on where in the input space a given instance sits. Their findings implied that two runs of LIME on the exact same prediction can produce meaningfully different explanations purely from the sampling noise, independent of any change to the underlying applicant. Zhou et al. (2021) treated this instability as a problem to be actively corrected rather than merely documented, proposing S-LIME, a modified version of LIME that uses a statistical hypothesis-testing procedure to determine how many perturbation samples are required before an explanation can be considered stable; their stated motivation was that LIME demonstrated instability posed a serious and practical threat to the users’ trust.

The most striking demonstration of this fragility comes from Slack et al. (2020), who showed that both LIME and SHAP can be deliberately manipulated by training a biased classifier to behave innocuously, specifically in the regions of the input space of LIME and SHAP sample when constructing an explanation. They generate some models that discriminate on a protected attribute while producing some explanations that hide this behaviour entirely. Although their attack scenario is adversarial and does not directly mirror the credit-scoring setting of this study, it establishes an important baseline finding, that an explanation method’s output can diverge sharply from a model’s true decision-making process under conditions that the analyst does not fully control, which is precisely the concern that this study investigates under ordinary, non-adversarial conditions, namely, whether small, natural perturbations to an applicant’s data produce disproportionately different explanations.

Taken together, this body of work has established that explanation stability is a well-documented concern in the general machine learning literature, with the proposed metrics (Alvarez-Melis & Jaakkola, 2018), documented failure modes (Zhang et al., 2019; Slack et al., 2020), and at least one proposed remedy specific to LIME (Zhou et al., 2021). What remains and is underexplored is how these findings translate to structured, tabular credit data specifically, where the features are fewer, more interpretable individually, and often correlated in ways particular to financial data such as income, credit amount, and external risk scores, and, more specifically, how explanation stability differs between an inherently interpretable baseline model and the black-box ensemble models increasingly favoured for their predictive performance. This is the gap that the present study addresses.

2.5 Model Choice: From Interpretable Baselines to Black-Box Ensembles

The tension between interpretability and predictive performance that motivates this study is reflected in the evolution of the underlying models themselves. Logistic regression remains the traditional, inherently interpretable baseline in credit scoring, valued precisely because its coefficients are directly and individually interpretable without recourse to a separate post-hoc explanation method. Breiman (2001) introduced Random Forests, an ensemble of decision trees combining bagging with random feature selection, which substantially improved predictive accuracy over single decision trees at the cost of the kind of direct, coefficient-level interpretability logistic regression offers. Chen and Guestrin (2016) introduced XGBoost, a scalable gradient boosting implementation that has since become one of the most widely used algorithms in applied credit-risk modelling, prized for its predictive performance, but even less directly interpretable than a Random Forest since its predictions emerge from a sequential, additive correction process across many boosted trees rather than a single averaging step. This progression, from a fully transparent linear model, through a bagged tree ensemble, to a boosted tree ensemble, forms a natural axis along which to study the central research questions of this study: whether the models that differ substantially in structural complexity, and therefore in their reliance on post-hoc explanation in the first place, also differ systematically in how stable their SHAP and LIME explanations are, and whether any gain in the predictive performance across that axis comes at a measurable cost in the explanation reliability.

2.6 Synthesis and Research Gap

The above reviewed literature supports three connected conclusions. First, there is a clear regulatory and practical case for explainability in credit scoring, specifically driven by data protection law (Goodman & Flaxman, 2017) and by a broader consensus in the finance literature that transparency is a precondition for trust in automated lending decisions (Černevičienė & Kabašinskas, 2024). Second, SHAP and LIME have become the dominant tools that researchers and practitioners used to meet that need, with a substantial and growing applied literature demonstrating their use in credit-scoring contexts (De Lange et al., 2022; Nallakaruppan et al., 2024; Busmann et al., 2021; Gramegna & Giudici, 2021). Third, and most consequentially for this study, a separate and largely independent strand of the wider machine learning literature has repeatedly shown that these same explanation methods are not inherently stable or reliable, and can vary substantially under the conditions that should not in principle change the resulting explanation (Alvarez-Melis & Jaakkola, 2018; Zhang et al., 2019; Zhou et al., 2021; Slack et al., 2020). These three strands of literature have rarely been brought together in a single, systematic study of credit scoring specifically. Much of the applied credit-scoring literature uses SHAP or LIME as though their outputs were self-evidently trustworthy, without directly testing whether similar applicants receive similar explanations, whether these explanations hold up under small, realistic perturbations to an applicant's data, or whether the choice between an interpretable baseline and a black-box ensemble carries a measurable stability cost alongside its predictive-performance benefit. This study addresses that gap directly, applying the stability concepts developed in the general interpretability literature, namely variance in feature attribution, rank consistency, and perturbation-based stability, to a real-world credit-scoring dataset, across both an interpretable baseline and more complex black-box models, in order to give the credit-scoring practitioners and regulators an evidence-based answer to a question the existing literature has largely assumed rather than tested.

Chapter 3: Data Preprocessing and Exploratory Data Analysis

The Home Credit Default Risk dataset is structured as a relational dataset made of several connected tables. The main table is `application_train` / `application_test`, where each row represents one current loan application. The training file contains the `TARGET` variable, which indicates whether the applicant experienced repayment difficulty, while the test file contains the same applicant information but without the `TARGET` column.

This dataset also includes several supporting tables containing historical credit behaviour. These supporting tables have a one-to-many relationship with the main application table, meaning one applicant can have multiple previous credit records, previous applications, or repayment records. Because of this, the supporting tables could not be merged directly with the application table, and were instead first aggregated to one row per applicant using `SK_ID_CURR`.

Four tables were used for this study: `application_train` / `application_test`, `bureau.csv`, `previous_application.csv`, and `installments_payments.csv`. These were selected because they provide interpretable and financially meaningful information about an applicant's current profile, external credit history, previous Home Credit application history, and actual repayment behaviour. Other tables, such as `bureau_balance.csv`, `POS_CASH_balance.csv`, and `credit_card_balance.csv`, were not used, since they are more granular monthly-balance tables that would significantly increase complexity and feature dimensionality, which could make SHAP and LIME explanations less clear and less stable. Since this study's focus is explanation stability rather than maximising leaderboard performance, a compact and interpretable feature set was preferred throughout.

This chapter is organized as follows. Section 3.1 covers preprocessing of the application table. Section 3.2 covers aggregation of the bureau table. Section 3.3 covers aggregation of the previous application table. Section 3.4 covers aggregation of the instalment table. Section 3.5 covers data quality issues after merging. Section 3.6 gives an overview of the merged dataset. Section 3.7 examines the target variable. Section 3.8 covers key numeric distributions. Section 3.9 covers missingness. Section 3.10 explores relationships with default risk. Section 3.11 presents the correlation structure. Section 3.12 closes the chapter with categorical feature patterns.

3.1 Application Table Preprocessing

The application table was preprocessed by transforming negative day-offset variables into positive, year-based features, creating financial ratio features to capture affordability more directly, and removing columns with low information value, high missingness, or weak interpretability. The same preprocessing logic was applied to both the `application_train` and `application_test` in order to keep the two files structurally consistent, aside from the `TARGET` column, which appears only in the training set.

3.1.1 Handling Invalid Employment Values

The `DAYS_EMPLOYED` column contained an anomalous placeholder value (365243) which does not represent a real employment duration. These rows correspond almost entirely to applicants whose `NAME_INCOME_TYPE` is Pensioner or Unemployed, a status already captured directly by that column. The anomalous value was first replaced with a missing value, converted into `EMPLOYMENT_YEARS` alongside the other date-offset variables, and then set to 0 to reflect that a current employment duration does not apply to this group. No separate missing-value indicator was added for this column, since `NAME_INCOME_TYPE` already identifies these applicants directly.

3.1.2 Transformation of Date-Offset Variables

Several time-related variables were originally stored as negative day offsets relative to the application date. These were transformed into positive year-based features to improve interpretability (as shown in Table 3.1), using $\text{Years Feature} = (-\text{Days Feature}) / 365$.

Table 3.1: Date-Offset Variable Transformations

Original Variable	Transformed Feature
<code>DAYS_BIRTH</code>	<code>AGE_YEARS</code>
<code>DAYS_EMPLOYED</code>	<code>EMPLOYMENT_YEARS</code>
<code>DAYS_REGISTRATION</code>	<code>REGISTRATION_YEARS</code>
<code>DAYS_ID_PUBLISH</code>	<code>YEARS_SINCE_ID_UPDATE</code>
<code>DAYS_LAST_PHONE_CHANGE</code>	<code>YEARS_SINCE_PHONE_CHANGE</code>

3.1.3 Financial Ratio Feature Engineering

Two ratio-based financial features were engineered to better represent affordability and repayment burden than raw monetary values alone. Table 3.2 shows how features were engineered.

Table 3.2: Engineered Financial Ratio Features

Engineered Feature	Formula
CREDIT_TO_INCOME_RATIO	$\text{AMT_CREDIT} / \text{AMT_INCOME_TOTAL}$
ANNUITY_TO_INCOME_RATIO	$\text{AMT_ANNUITY} / \text{AMT_INCOME_TOTAL}$

3.1.4 Column Removal

After transformation into year-based variables, the original negative day-offset columns (DAYS_BIRTH, DAYS_EMPLOYED, DAYS_REGISTRATION, DAYS_ID_PUBLISH, DAYS_LAST_PHONE_CHANGE) were removed. Several further groups of columns were also removed to reduce noise and improve interpretability:

- Low-information contact features (FLAG_MOBIL, FLAG_CONT_MOBILE, FLAG_WORK_PHONE) were removed for near-zero variance or redundancy. FLAG_EMP_PHONE, FLAG_PHONE, and FLAG_EMAIL were retained as potentially meaningful behavioural signals.
- All FLAG_DOCUMENT_* variables were removed, being highly sparse and weakly interpretable.
- Building and housing metadata (APARTMENTS_*, BASEMENTAREA_*, YEARS_BUILD_*, COMMONAREA_*, ELEVATORS_*, LIVINGAREA_*, NONLIVINGAREA_*, WALLSMATERIAL_*, EMERGENCYSTATE_*) were removed for substantial missingness, weak interpretability, and correlated feature groups.

The Demographic, financial, employment and occupation, external risk score (EXT_SOURCE_1-3), geographic and regional risk, social-circle default, credit bureau enquiry, and application timing feature columns, were all retained, as each provides an interpretable and financially meaningful signal.

3.2 Bureau Table Preprocessing and Aggregation

The bureau.csv table contains multiple previous credit records for the same applicant, and was aggregated by SK_ID_CURR to create one applicant-level credit history table (Table 3.3). CREDIT_CURRENCY was removed, adding limited interpretability and unnecessary categorical complexity. The following features were then derived using count, sum, maximum, and minimum operations grouped by SK_ID_CURR, selected because they directly describe the previous borrowing activity, outstanding debt, overdue behaviour, and credit type in an interpretable way.

Table 3.3: Aggregated Bureau Features

Aggregated Feature	Formula / Derivation	Meaning
TOTAL_PREVIOUS_LOANS	Count of SK_ID_BUREAU per SK_ID_CURR	Total number of previous bureau credit records for the applicant
ACTIVE_PREVIOUS_LOANS	Count of CREDIT_ACTIVE = "Active" per SK_ID_CURR	Number of previous bureau credits still active
TOTAL_BUREAU_DEBT	Sum of AMT_CREDIT_SUM_DEBT per SK_ID_CURR	Total outstanding debt from previous bureau credits
TOTAL_OVERDUE_DEBT	Sum of AMT_CREDIT_SUM_OVERDUE per SK_ID_CURR	Total overdue debt from previous bureau credits

MAX_DAYS_OVERDUE	Max of CREDIT_DAY_OVERDUE per SK_ID_CURR	Worst recorded overdue delay in previous bureau credits
MOST_RECENT_PREVIOUS_CREDIT_YEARS	Minimum of (-DAYS_CREDIT / 365) per SK_ID_CURR	How recently the applicant took a previous bureau credit
COUNT_CREDIT_CARDS	Count of CREDIT_TYPE = "Credit card" per SK_ID_CURR	Number of previous credit card records
COUNT_CONSUMER_CREDITS	Count of CREDIT_TYPE = "Consumer credit" per SK_ID_CURR	Number of previous consumer credit records

3.3 Previous Application Table Preprocessing and Aggregation

The previous_application.csv table contains multiple previous Home Credit application records for the same applicant, and was aggregated by SK_ID_CURR to create one applicant-level history table (as shown in Table 3.4). Several columns were dropped for high missingness or unnecessary complexity, including interest-rate variables, down-payment variables, repayment-schedule dates, NAME_TYPE_SUITE, and the insurance indicator. The final aggregation was based on the most relevant and reliable remaining columns: SK_ID_CURR, SK_ID_PREV, NAME_CONTRACT_STATUS, AMT_APPLICATION, AMT_CREDIT, and DAYS_DECISION.

Table 3.4: Aggregated Previous Application Features

Aggregated Feature	Formula / Derivation	Meaning
TOTAL_PREVIOUS_APPLICATIONS	Count of SK_ID_PREV per SK_ID_CURR	Total number of previous Home Credit applications made by the applicant
APPROVED_PREVIOUS_APPLICATIONS	Count of NAME_CONTRACT_STATUS = "Approved" per SK_ID_CURR	Number of previous applications that were approved
REFUSED_PREVIOUS_APPLICATIONS	Count of NAME_CONTRACT_STATUS = "Refused" per SK_ID_CURR	Number of previous applications that were refused
PREVIOUS_APPROVAL_RATE	APPROVED_PREVIOUS_APPLICATIONS / TOTAL_PREVIOUS_APPLICATIONS	Proportion of previous applications that were approved
PREVIOUS_REFUSAL_RATE	REFUSED_PREVIOUS_APPLICATIONS / TOTAL_PREVIOUS_APPLICATIONS	Proportion of previous applications that were refused
AVG_PREVIOUS_APPLICATION_AMOUNT	Mean of AMT_APPLICATION per SK_ID_CURR	Average amount the applicant previously applied for
AVG_PREVIOUS_CREDIT_AMOUNT	Mean of AMT_CREDIT per SK_ID_CURR	Average credit amount previously granted
MAX_PREVIOUS_CREDIT_AMOUNT	Max of AMT_CREDIT per SK_ID_CURR	Largest previous credit amount granted to the applicant
MOST_RECENT_PREVIOUS_APPLICATION_YEARS	Minimum of (-DAYS_DECISION / 365) per SK_ID_CURR	How recently the applicant had a previous Home Credit application

The missing values after aggregation were filled with 0, so the applicants with no recorded value for a given feature could remain in the modelling dataset. The Amount-based features were rounded to two decimal places, the approval and refusal rates were rounded to three decimal places, and the MOST_RECENT_PREVIOUS_APPLICATION_YEARS was rounded to one decimal place for readability purpose.

3.4 Installment Table Preprocessing and Aggregation

The installments_payments.csv table was used to capture actual repayment behaviour, and was aggregated by SK_ID_CURR (as shown in Table 3.5). Payment delay was calculated as $\text{PAYMENT_DELAY_DAYS} = \text{DAYS_ENTRY_PAYMENT} - \text{DAYS_INSTALMENT}$, then converted into $\text{LATE_PAYMENT_DAYS} = \max(\text{PAYMENT_DELAY_DAYS}, 0)$, so that the early and on-time payments count as 0 while the late payments retain their positive delay values. This ensures the final the delay features only measure the late repayment behaviour.

Table 3.5: Aggregated Instalment Features

Feature	Formula / Derivation	Meaning
TOTAL_INSTALLMENT_RECORDS	Count of SK_ID_PREV per SK_ID_CURR	Total number of installment/payment records available for the applicant
LATE_PAYMENT_COUNT	Count of records where $(\text{DAYS_ENTRY_PAYMENT} - \text{DAYS_INSTALMENT}) > 0$	Number of installment payments made after the scheduled due date
LATE_PAYMENT_RATE	$\text{LATE_PAYMENT_COUNT} / \text{TOTAL_INSTALLMENT_RECORDS}$	Proportion of the applicant's installment payments that were late
AVG_LATE_PAYMENT_DAYS	Mean of $\max(\text{DAYS_ENTRY_PAYMENT} - \text{DAYS_INSTALMENT}, 0)$ per SK_ID_CURR	Average late-payment delay in days. Early and on-time payments count as 0
MAX_LATE_PAYMENT_DAYS	Max of $\max(\text{DAYS_ENTRY_PAYMENT} - \text{DAYS_INSTALMENT}, 0)$ per SK_ID_CURR	Worst/longest late-payment delay recorded. Early and on-time payments count as 0

Missing values after aggregation were filled with 0. LATE_PAYMENT_RATE was rounded to three decimal places, and AVG_LATE_PAYMENT_DAYS and MAX_LATE_PAYMENT_DAYS to two decimal places.

3.5 Merging and Post-Merge Data Quality

After the data preprocessing and aggregation, all supporting tables were merged into the processed application_train and application_test datasets using a left join on SK_ID_CURR, so that every applicant in the main application table was retained even without records in one of the supporting tables. The missing values created by this merge were filled with 0 only for the newly merged bureau, previous application, and installment features, indicating no recorded history for that applicant. The Application-level fields that were already missing before merging, such as the external risk scores and vehicle age, were handled separately below, so that a genuine missingness was not confused with an applicant simply having no supporting-table history.

3.5.1 Restoration and Imputation of External Risk Scores

EXT_SOURCE_1, EXT_SOURCE_2, and EXT_SOURCE_3 are external credit bureau scores with differing levels of missingness (56.4%, 0.2%, and 19.8% respectively). The missing values were restored to genuine NaN after merging. EXT_SOURCE_2, missing for only a small fraction of applicants, was filled with the training-set median. EXT_SOURCE_1 and EXT_SOURCE_3 are missing for a much larger share of applicants, so filling them the same way would repeat a single value across a large portion of the dataset, creating an artificial spike that could distort both model training and later the SHAP and LIME explanations for those applicants.

Applicant age, income, credit amount, and the other EXT_SOURCE scores were checked for their relationship with EXT_SOURCE_1 and EXT_SOURCE_3 before choosing an imputation approach; EXT_SOURCE_1, for

example, correlates at 0.60 with applicant age, a substantial relationship. Missing values in EXT_SOURCE_3 and EXT_SOURCE_1 were therefore imputed using a linear regression model trained on the applicants with a known score, using applicant age, income, credit amount, employment years, and, for EXT_SOURCE_1, the completed EXT_SOURCE_2 and EXT_SOURCE_3 values as predictors. EXT_SOURCE_3 was imputed before EXT_SOURCE_1 specifically, so its completed values could then serve as an additional predictor for EXT_SOURCE_1, the more heavily missing of the two. This model, fitted only on the training set, was used to fill missing values in both the training and test sets. A binary indicator column was added for each of the three features to preserve the information that a value had originally been missing.

The predictors used explain only a modest share of the variation in EXT_SOURCE_1 and EXT_SOURCE_3. The resulting imputed values are a closer and more individualized estimate than a single repeated median, but they remain an estimate rather than a substitute for the real, unavailable external score. This limitation of these two features is worth keeping in mind during the modelling and explanation analysis phases.

3.5.2 Handling of Not-Applicable Vehicle Age

OWN_CAR_AGE does not apply to applicants who do not own a car, a status already captured by FLAG_OWN_CAR. For these applicants, the field was left as a missing value rather than filled with 0 since a value of 0 would incorrectly imply a car that has 0 as age. No separate indicator column was added, since FLAG_OWN_CAR already identifies these applicants directly. A small number of car-owning applicants with a genuinely unrecorded car age were also categorized as missing.

3.5.3 Handling of Extreme Income Values

One applicant in the training data recorded an AMT_INCOME_TOTAL of 117,000,000, more than six times higher than the next highest value in the dataset and inconsistent with the smooth, gradual distribution of all other income values. This was treated as a data entry error, and the row was removed from the training data. No comparable extreme value was present in the test data, so no rows were removed there.

3.5.4 Credit-to-Goods Price Ratio

AMT_CREDIT and AMT_GOODS_PRICE were found to be highly correlated ($r = 0.987$), providing largely duplicate information to downstream models. AMT_GOODS_PRICE was replaced with a new engineered feature, CREDIT_GOODS_RATIO = AMT_CREDIT / AMT_GOODS_PRICE, retaining the meaningful relationship between the two variables without redundancy. A small number of applicants with a missing AMT_GOODS_PRICE value were assigned the median ratio calculated from the training data.

3.5.5 Encoding of High-Cardinality Categorical Features

ORGANIZATION_TYPE (58 categories) and OCCUPATION_TYPE (18 categories, including an Unknown category for previously missing values) were each converted into a single numeric feature using smoothed target encoding (Micci-Barreca, 2001), where a category is replaced with its average TARGET rate, weighted toward the overall training-set average in proportion to how few applicants belonged to that

$$\text{smoothed rate} = (n \times \text{category rate} + k \times \text{overall rate}) / (n + k)$$

where n is the number of applicants in the category, k is a smoothing weight, and the overall rate is the average TARGET rate across the training data. This prevents small categories, some with fewer than 100 applicants, from producing unstable, overconfident estimates, while avoiding the large increase in feature count that one-hot encoding both variables would have caused. Encoding rates were calculated using only the training data and applied consistently to the test data; categories present only in the test data were assigned the overall training-set average rate. A smoothing weight of $k = 50$ was used throughout, and tested against $k = 20$ and $k = 100$ using validation-set ROC-AUC; the difference between the best and worst result was 0.00012, confirming the choice of k is not a meaningful source of variation within a reasonable range.

3.5.6 Encoding of the Remaining Categorical Features

A further 10 application-level columns were converted from text to numeric form, each treated according to its own category count and relationship with TARGET. NAME_CONTRACT_TYPE, FLAG_OWN_CAR, and FLAG_OWN_REALTY each have two categories and were mapped directly to a single binary column. CODE_GENDER, with three categories including a small XNA group of four applicants, was one-hot encoded into three indicator columns. NAME_EDUCATION_TYPE was encoded as a single ordinal column (0 to 4),

reflecting its natural order from Lower secondary through to Academic degree. NAME_TYPE_SUITE, NAME_INCOME_TYPE, NAME_FAMILY_STATUS, and NAME_HOUSING_TYPE were each converted using the same smoothed target encoding described above, with NAME_TYPE_SUITE's missing values grouped into an explicit Unknown category first. WEEKDAY_APPR_PROCESS_START was removed, since default rate varied by only 0.59 percentage points across all seven days of the week.

3.5.7 Bureau Enquiry and Social Circle Missing Values

Two further groups of columns were missing together across every column in the group, for the same applicants. The six AMT_REQ_CREDIT_BUREAU_* columns, recording bureau enquiries across different time windows, are missing together for 13.50% of applicants, all of whom also have TOTAL_PREVIOUS_LOANS equal to 0, meaning no bureau history at all. These were filled with 0, a true value rather than an estimate, with no separate indicator needed since TOTAL_PREVIOUS_LOANS already identifies this group.

OBS_30_CNT_SOCIAL_CIRCLE, DEF_30_CNT_SOCIAL_CIRCLE, OBS_60_CNT_SOCIAL_CIRCLE, and DEF_60_CNT_SOCIAL_CIRCLE, recording payment difficulties within an applicant's social circle, are missing together for 0.33% of applicants, with no other column explaining the pattern. Each was filled with its training-set median, alongside a single shared indicator column, SOCIAL_CIRCLE_MISSING. A small number of remaining columns, AMT_ANNUITY (12 rows), CNT_FAM_MEMBERS (2 rows), YEARS_SINCE_PHONE_CHANGE (1 row), and ANNUITY_TO_INCOME_RATIO (12 rows, inherited from AMT_ANNUITY), were filled with their training-set median, with no indicator column given how few rows were involved.

3.6 Dataset Overview

The final dataset brings together the four source tables described above into a single row per applicant, joined on SK_ID_CURR. The final merge is visualized in Figure 3.1.

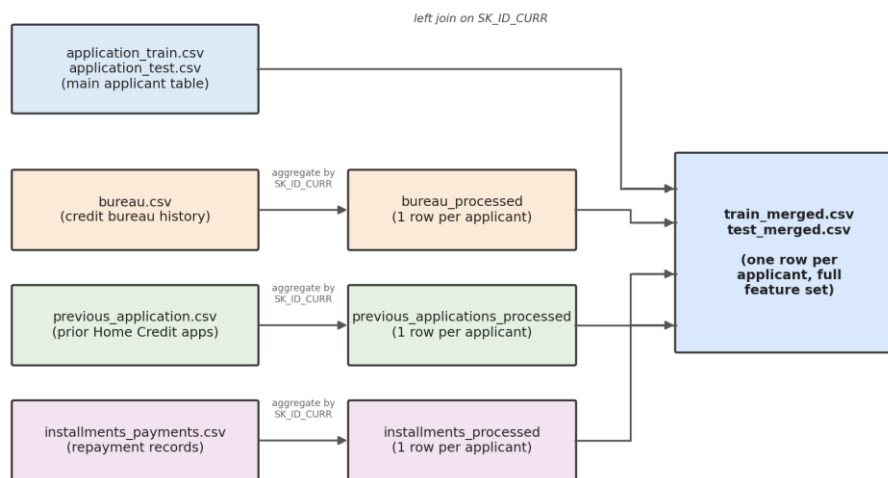


Figure 3.1: How the four source tables combine into the final merged dataset.

As represented in Figure 3.2, not every applicant has a record in every supporting table, and coverage varies: 85.69% of applicants have at least one bureau record, 94.65% have at least one previous Home Credit application, and 94.84% have at least one installment record. 80.88% have history in all three tables, while only 0.73% have no history in any of them, applicants who are new to both Home Credit and, based on the available data, to formal credit more broadly.

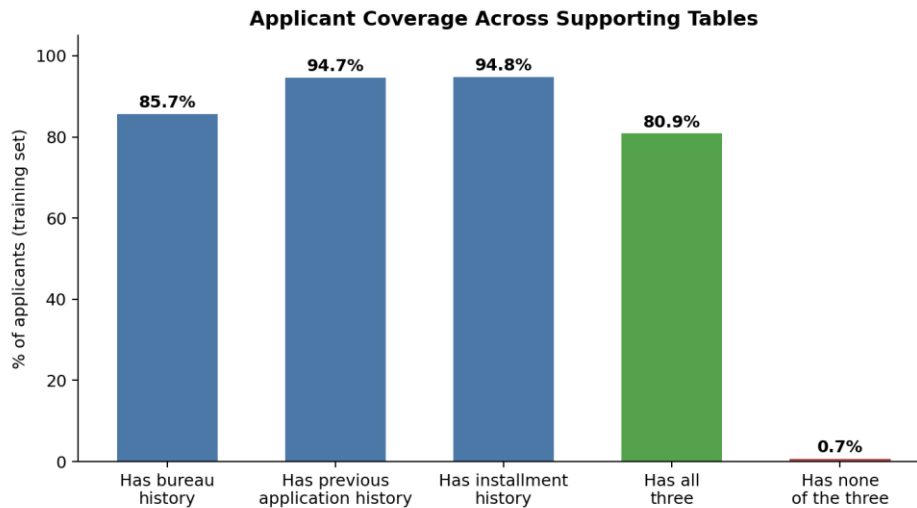


Figure 3.2: Share of training applicants with a record in each supporting table.

3.7 Target Variable

TARGET marks whether an applicant experienced repayment difficulty. The training data is heavily imbalanced toward non-default, an overall baseline rate of 8.07% (as shown in Figure 3.3), which shapes both the modelling approach described in Chapter 4 and how the default-rate comparisons should be read throughout this chapter: any category rate meaningfully above or below that figure is notable.

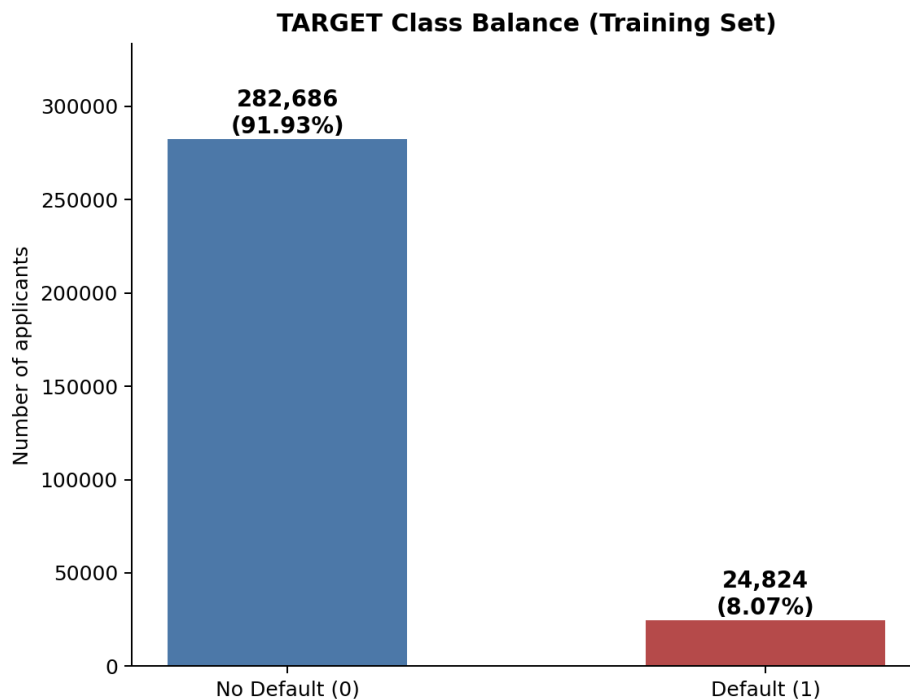


Figure 3.3: TARGET distribution across the training set.

3.8 Key Numeric Distributions

AMT_INCOME_TOTAL, AMT_CREDIT, and AGE_YEARS in Figure 3.4 all show the shapes typical of financial and demographic data: income and credit amount are right-skewed, with most applicants clustered at the lower end and a long tail of larger values, while age is closer to a broad, gently peaked spread across the working adult population. CREDIT_GOODS_RATIO, the ratio of credit amount received to the price of goods purchased, clusters tightly around 1.0, most applicants borrow close to the price of the goods they are financing, with a visible spread of the applicants borrowing notably more.

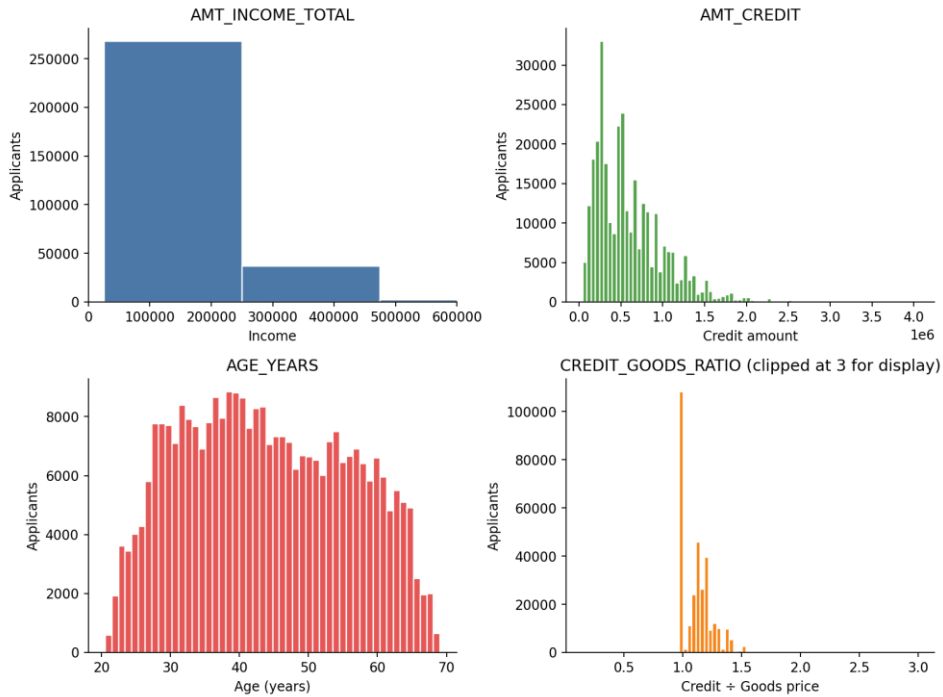


Figure 3.4: Distributions of income, credit amount, age, and the credit-to-goods ratio.

The three external risk scores are each roughly bell-shaped across their 0-1 range (Figure 3.5), though EXT_SOURCE_1 and EXT_SOURCE_3 show some smoothing around their central values, a visible trace of the predictive imputation applied to their originally-missing values.

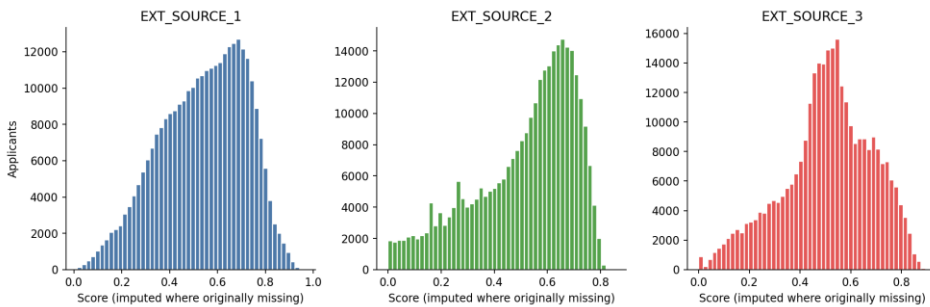


Figure 3.5: Distributions of the three external risk scores.

3.9 Missingness Overview

Before any imputation, several columns had substantial genuine missingness. OWN_CAR_AGE and EXT_SOURCE_1 stand out as the most incomplete, followed by EXT_SOURCE_3 and the credit bureau enquiry columns. Most other columns are essentially complete, with missingness below half a percent (as illustrated in Figure 3.6).

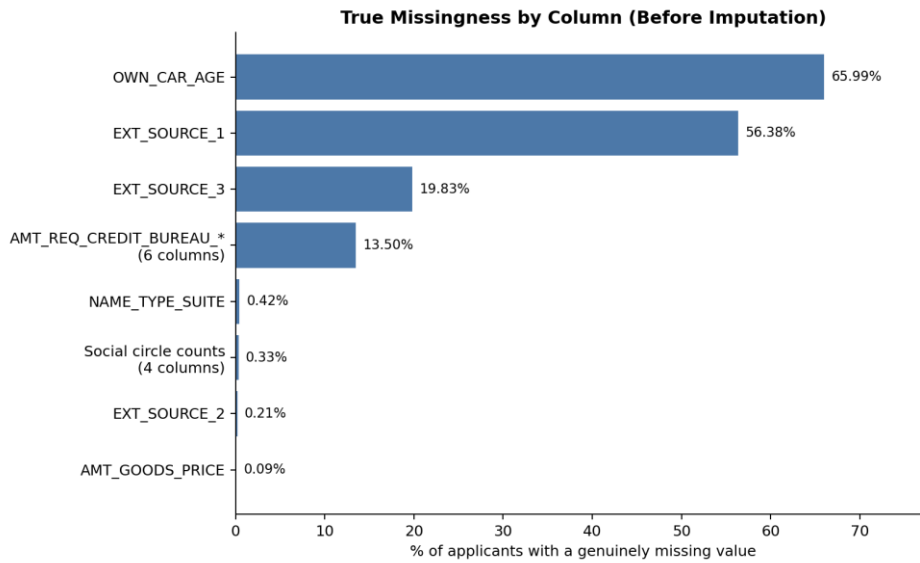


Figure 3.6: Share of applicants with a genuinely missing value, by column, before imputation.

3.10 Relationships With Default Risk

As Figure 3.7 shows, `EXT_SOURCE_1` and `EXT_SOURCE_2` both correlate with `TARGET` individually. Binning applicants by both scores and calculating the real default rate within each bin shows a clear, substantial gradient: default rate ranges from 36.6% for applicants with the lowest `EXT_SOURCE_1` and `EXT_SOURCE_2` scores down to 1.2% for those with the highest scores on both, a genuine and fairly strong relationship, consistent with these two features being among the strongest correlates of `TARGET` in the dataset, shown in the next section.

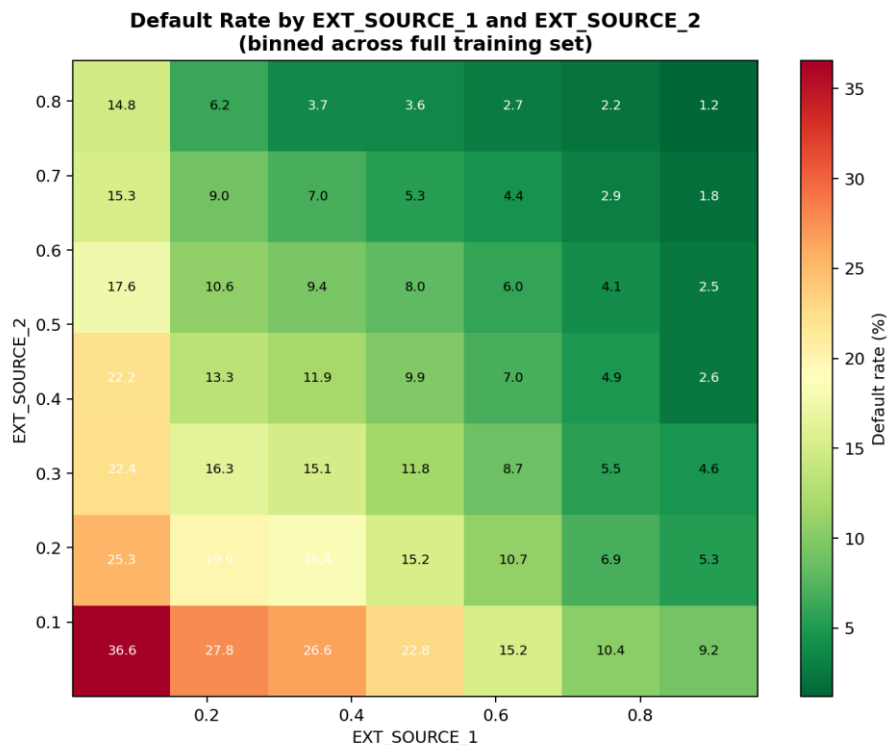


Figure 3.7: Default rate by `EXT_SOURCE_1` and `EXT_SOURCE_2`, binned across the full training set.

Default risk also declines steadily with age, from 11.43% in the 20-30 bracket down to 4.93% in the 60-70 bracket, a gap of more than double between the youngest and oldest groups (as shown in Figure 3.8).

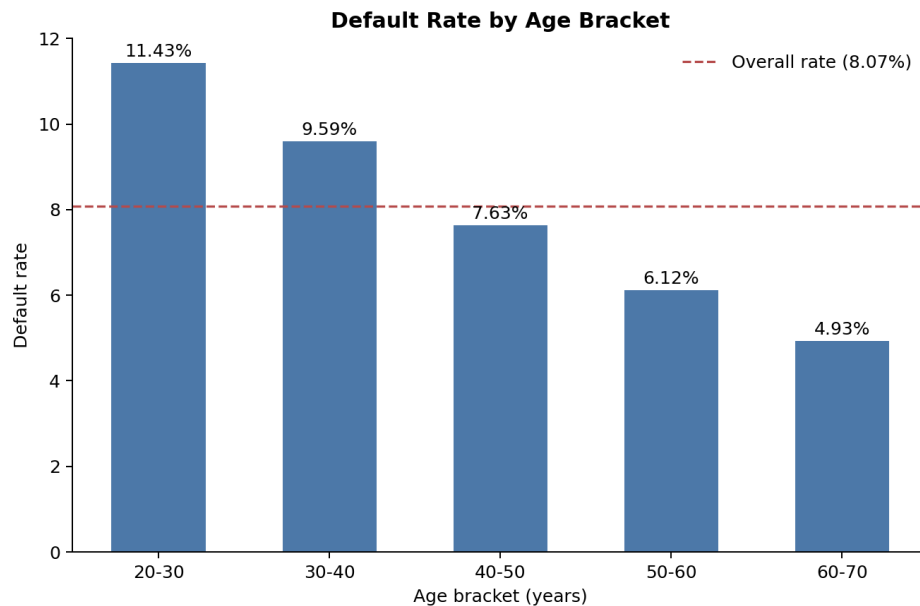


Figure 3.8: Default rate by age bracket.

Missingness itself also relates to default risk, not always in the same direction (as shown in Figure 3.9). For EXT_SOURCE_1 and EXT_SOURCE_3, applicants missing a score default somewhat more often than those with one present (8.52% vs. 7.49%, and 9.31% vs. 7.77%, respectively). Social circle counts and occupation type show the opposite pattern, applicants missing this information default less often (3.53% vs. 8.09%, and 6.51% vs. 8.79%), most likely reflecting that a large share of this group are retirees, who default less overall regardless of the missing data itself. This is part of the reasoning behind preserving missingness as its own signal, the *_MISSING flag columns, rather than only filling the gaps.

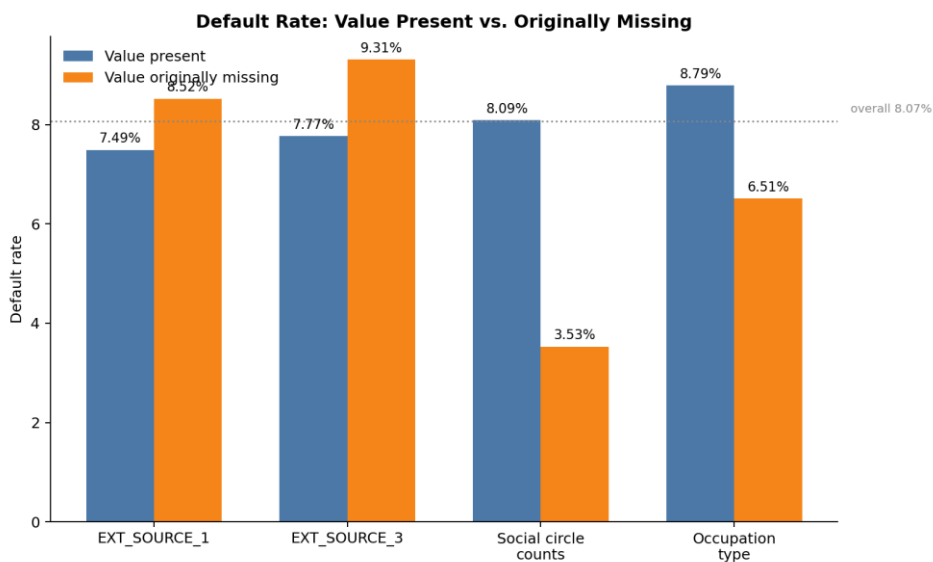


Figure 3.9: Default rate for applicants with a value present vs. originally missing, by feature.

3.11 Correlation Structure

Figure 3.10 below shows that across the engineered feature set, EXT_SOURCE_1/2/3 are the strongest individual correlates of TARGET (-0.14 to -0.16), consistent with the patterns shown above, and are likely to be influential features once models are trained. CREDIT_GOODS_RATIO (0.069) and LATE_PAYMENT_RATE (0.070) are the next-strongest, ahead of the raw AMT_CREDIT and AMT_INCOME_TOTAL amounts on their own, some evidence that the engineered ratio and behavioural features carry more signal than the raw monetary values did individually. No pair of features in this view shows the kind of extreme correlation seen between the original AMT_CREDIT and AMT_GOODS_PRICE, addressed earlier in this chapter; the strongest

relationships here are the expected ones, such as CREDIT_TO_INCOME_RATIO and ANNUITY_TO_INCOME_RATIO both deriving from AMT_INCOME_TOTAL.

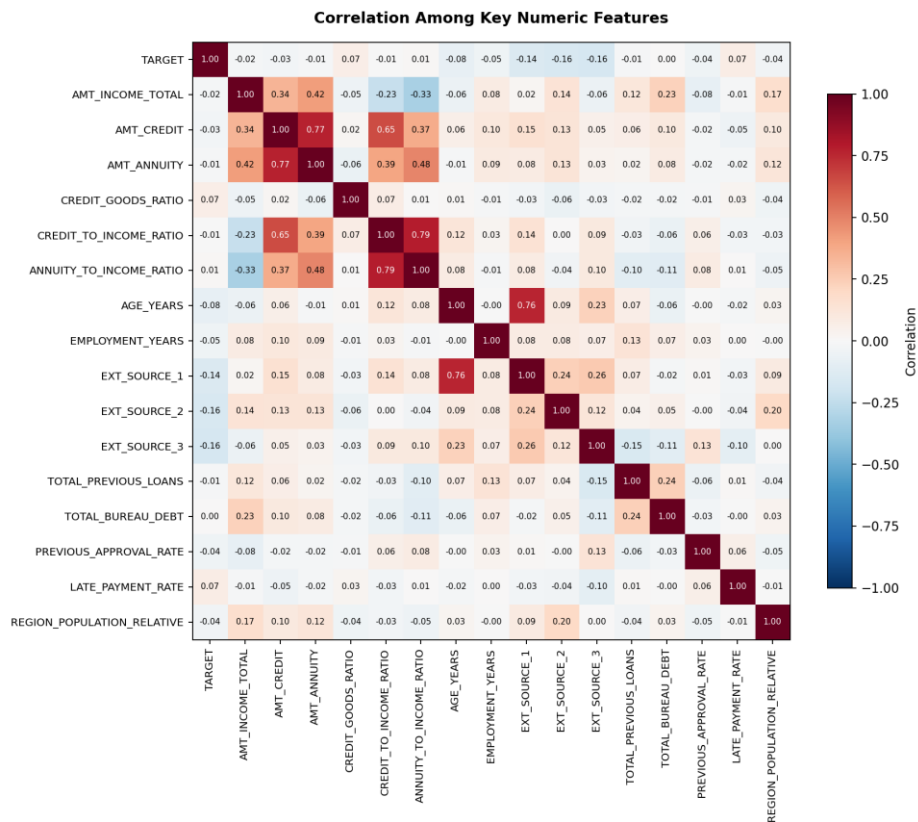


Figure 3.10: Correlation matrix across key numeric features, including TARGET.

3.12 Categorical Feature Patterns

Education level shows a clear and ordered relationship with the default risk, the reasoning behind encoding it as an ordinal feature rather than one-hot: default rate falls steadily from 10.93% for applicants with Lower secondary education to just 1.83% for those with an Academic degree. Housing situation also separates applicants meaningfully, from 6.57% (Office apartment) to 12.31% (Rented apartment), owning or living in a stable arrangement associates with lower risk than renting or living with parents. Among the largest organization types, risk ranges from 5.40% (the largely-retired XNA group) up to 10.17% (Self-employed), reflecting the kind of category-level variation that motivated target-encoding these features rather than treating them as uninformative. Figure 3.11 illustrates below these patterns.

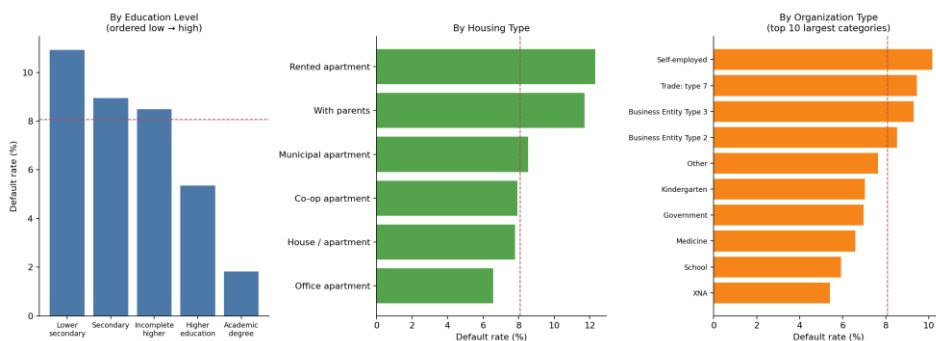


Figure 3.11: Default rate by education level, housing type, and top organization types.

Chapter 4: Methodology

The aim of this study is to evaluate the stability and consistency of explainable AI methods in credit scoring, and to understand how that stability relates to predictive performance across models of differing complexity. This chapter sets out the methodology built to achieve that aim: preparing a defensible feature set from the merged dataset, developing four credit scoring models that span simple to complex architectures, generating SHAP and LIME explanations for those models in a form that allows fair comparison, and designing the stability tests that answer the four research questions.

The chapter follows the order in which the pipeline was actually built. Section 4.1 covers the features selection. Section 4.2 covers the model development. Section 4.3 covers the explanation framework. Section 4.4 covers the stability testing methodology.

4.1 Feature Selection

Before building the models in this study, the 79-column dataset described in the previous chapter needed to be reduced to a final feature set. Not every column in that dataset was necessarily useful for prediction, and every column that was kept would later need to be explained by SHAP and LIME. This made feature selection more important for this study than it would be for a project focused on prediction alone: a weak or redundant feature can lower model performance, and it can also show up in an explanation as noise.

4.1.1 Validation Strategy

An 80/20 stratified split was taken from the training data before any selection decision was made. This preserved the 8.07% default rate identically in both resulting pieces. The resulting validation set, fixed with a constant random seed, was reused for every decision covered in this chapter, including feature selection, model evaluation, and threshold calibration. This meant no methodological choice was ever validated against the data it would later be judged on. The true test set was not touched at any point during the model development.

4.1.2 Selection Philosophy

A conservative approach was taken for features selection. A feature was kept by default and only removed for a specific reason, either for the reason that it was effectively constant across applicants or it was clearly redundant with another feature already being kept. The correlation with the TARGET variable was not used as a removal criterion. Random Forest and XGBoost can pick up on patterns and interactions between the features that a simple correlation score would miss, so filtering on correlation alone would have removed the features that mattered to the more complex models, but not to the interpretable baseline, making the later model comparison unfair.

4.1.3 Near-Constant and Redundancy Screening

Every feature was checked for how much variation it had. One gender indicator showed almost none, affecting fewer than three applicants out of 246,008, and was removed. The remaining features were checked for redundancy using correlation between pairs of features, not between a feature and TARGET. Six near-duplicate features were found this way: two gender indicators that became mirror images of each other once the rare category above was removed, two near-identical social-circle counts, and a regional risk rating superseded by a more detailed version. For each pair, the feature more strongly correlated with TARGET was kept. The pairs that moved together without being near duplicates, such as family size and number of children, were left untouched. This screening reduced the feature set from 79 to 73 columns.

4.1.4 Validating the Engineered Aggregate Features

The bureau, previous application, and instalment features built during preprocessing were checked once more at this stage. The goal was to confirm that the aggregation had produced sensible values, and each feature showed real variation among the applicants it applied to, not just among the larger group with no history in that table. All rate-based features fell within the expected 0 to 1 range, no applicant had more approved and refused applications than their real total, and no values were negative where they shouldn't be. Looking only at applicants with real history in each table, every feature showed a genuine spread of values rather than sitting close to one number. No feature was removed at this stage since the purpose here was to assess and confirm the data quality rather than selecting the features.

4.1.5 Finalizing the Feature Set

One column, `OWN_CAR_AGE`, was left with genuine missing values during preprocessing. Its correct treatment depended on how the different models would use it. Three of the four models require a value in every cell, while only XGBoost can handle missing values directly, and giving each model a different version of the same feature would have made the later model comparison unfair. Two different situations existed within this column. Most of the missing values belonged to the applicants with no car at all, for whom the field simply does not apply. These were filled with 0. It is safe to do so because `FLAG_OWN_CAR` sits alongside `OWN_CAR_AGE` in every model's input and already states that the applicant has no car, so the value is never read on its own. A much smaller group, five applicants across the training data, were car owners whose age was never recorded. These were filled with the median car age among applicants who do own one, giving a real estimate for a genuine unknown rather than a placeholder. This produced a final, complete 73-column features set, used identically across all the four models, for the rest of this study.

4.2 Predictive Model Development

With the final features set in place, four credit scoring models were developed, namely: Logistic Regression as the interpretable baseline, single Decision Tree, Random Forest as a bagged ensemble, and XGBoost as ensemble model. These models were chosen to cover a real range of structural complexity, from a fully transparent linear model through to a more complex boosted ensemble.

4.2.1 Class Imbalance Handling

`TARGET` was imbalanced at 91.9% non-default versus 8.1% default, and this was addressed through class weighting. Each defaulter row was weighted about 11.4 times more heavily than a non-defaulter row during training. This number comes directly from the class split. Dividing the number of non-defaulters by the number of defaulters in the training data, 282,686 divided by 24,825, gives a ratio of roughly 11.4.

4.2.2 Hyperparameter Settings

Each model used standard widely accepted hyperparameters setup. The contribution of this study is in explanation stability, not in the maximum predictive performance, and heavy tuning risked shifting focus away from the research questions and toward chasing better looking numbers. Where a setting genuinely needed adjusting (this is covered in the next two sections), the change was made once for a specific reason, and then left fixed.

4.2.3 Logistic Regression: Feature Scaling

An early run of Logistic Regression using the full 73 features set failed to converge. The cause was a mismatch in scale across the features. Applicant income is measured in the hundreds of thousands, while the `EXT_SOURCE` scores sit between 0 and 1. Logistic Regression is known to struggle when features sit on very different scales like this. Tree based models do not have this problem. The fix was to rescale every feature before training by subtracting each feature's average and dividing by its typical spread. This brings every feature onto a comparable scale, centred around 0, regardless of its original units. This rescaling was calculated using training data only and then applied to the validation and test sets without being recalculated.

4.2.4 Random Forest: Decision Threshold Calibration

A separate issue came up with Random Forest. Using the standard 0.5 threshold to classify an applicant as high risk, the model's predicted probabilities were heavily compressed toward the low end of the 0 to 1 range. This happens because Random Forest averages predictions across 100 trees, which dilutes any single tree's confidence more than a single, unaveraged model would. Diagnostic inspection confirmed that this compression did not affect the model's threshold-independent ranking ability. The problem was that its naturally compressed output range no longer lined up with a standard 0.5 cutoff. To fix this, the decision threshold for Random Forest was recalibrated to the real training set default rate of 8.07%, a single adjustment based directly on the data rather than a search for the best-looking cutoff. Logistic Regression, Decision Tree, and XGBoost did not show this compression and kept the standard 0.5 threshold.

4.2.5 Models Evaluation

All four models were evaluated on the same validation set using accuracy, precision, recall, F1-score, and ROC-AUC as metrics. ROC-AUC was utilized as the main basis for comparing the models since it does not depend

on a chosen decision threshold. The other considered metrics do, and the Random Forest's threshold differs from the other three models, so it would not be an accurate comparison. The other metrics are still reported alongside ROC-AUC for a full picture of each model's performance.

4.3 Explainability Framework

With the four trained models in place, the next step was to generate the SHAP and LIME explanations needed to test how stable those explanations are. SHAP and LIME were chosen because they are model agnostic and are already the standard choice for explaining credit scoring predictions in the literature already reviewed. That same literature has also raised real concerns about how stable these methods actually are, concerns that have rarely been tested on structured financial data like this dataset.

4.3.1 Applicant Sampling Strategy

Generating an explanation for every one of the 61,502 validation applicants, across the four models and two explanation methods, was not computationally practical. A fixed sample of 200 applicants was therefore drawn at random from the validation set (constant seed) and reused identically across every model and every explanation method in this study, ensuring that any later comparison across models or methods was always evaluated against the same underlying applicants.

4.3.2 Ensuring Comparable Explanation Units Across Models

Different models calculate their raw internal output differently. Logistic Regression produces an unbounded internal score before it is converted into a probability, while tree-based models work in log odds by default. If SHAP were left to read each model's own internal scale, the resulting values would not be measured in the same units across models, similar to comparing a temperature in Celsius directly against one in Kelvin without converting first. To keep every model on the same footing, each SHAP explainer was built around a model's `predict_proba` output specifically, the function that returns an actual probability between 0 and 1. This meant every model, regardless of its type, reported feature contributions in the same unit: percentage points of predicted default risk. This approach was verified before being used throughout the study. Feature rankings for a given model stayed consistent regardless of which output SHAP was pointed at, and the largest contribution seen in testing fell within the expected 0 to 1 range.

4.3.3 SHAP Implementation

Each SHAP explainer was built using this `predict_proba` based approach, with a background reference of 100 applicants sampled at random from the training data using a fixed seed. This background sample gives SHAP a baseline for what a typical applicant looks like, which each explained applicant's prediction is then compared against. A sample of 100 is standard practice, large enough for a stable baseline without being computationally expensive. The same 100 applicants were used as the background sample for every model, keeping this part of the comparison consistent throughout.

4.3.4 LIME Implementation

LIME was configured with continuous-feature discretisation disabled, so that every explanation returned one clean numeric weight per feature rather than LIME's default behaviour of grouping continuous features into readable ranges. This was necessary to produce output in the same per-feature, per-applicant vector format as the SHAP explanations, which the stability metrics defined below require. Unlike SHAP, LIME's explanations are already expressed directly in terms of the model's probability output by construction, so no equivalent unit-correction step was needed for LIME.

4.4 Explanation Stability

The Explanation stability was tested using two independent experiments: a controlled perturbation test and a real-world similarity test, each targeting a different, complementary notion of stability, before their results were brought together for the cross-model and performance-stability comparisons that follow. The formulas used throughout the rest of this study are defined as follows.

4.4.1 Perturbation-Based Stability

This test measures whether an individual applicant's explanation stays consistent when their own data changes slightly. A subset of 50 applicants was drawn from the original 200 applicant sample, and five perturbed

versions of each were generated, giving 250 perturbed instances in total. Each version applied an independent random adjustment of up to five percent, in either direction, to every genuinely continuous feature in an applicant's profile: monetary amounts, ages, and continuous scores or rates. The categorical, ordinal, binary, and count based features were left untouched since a small percentage change has no realistic meaning for a flag, a small integer count, or a category.

Two engineered ratio features: CREDIT_TO_INCOME_RATIO and ANNUITY_TO_INCOME_RATIO, are calculated directly from other features being perturbed independently (AMT_CREDIT, AMT_INCOME_TOTAL, and AMT_ANNUITY). Perturbing the ratio and its components separately would have produced an applicant whose numbers no longer agreed with each other, so these two ratios were recalculated from their own newly perturbed components after each perturbation instead. CREDIT_GOODS_RATIO could not be handled the same way, since its original source feature, AMT_GOODS_PRICE, was engineered out of the dataset during preprocessing. It was perturbed as an independent value instead.

Fresh SHAP and LIME explanations were generated for all 250 perturbed instances using the same procedure described above. Each perturbed applicant's explanation was then compared back against that same applicant's original explanation using cosine similarity, consistent with the approach used to test explanation robustness in Alvarez-Melis and Jaakkola (2018):

$$Stab(x, x') = \frac{e(x) \cdot e(x')}{\|e(x)\| \|e(x')\|}$$

where $e(x)$ is the original explanation vector and $e(x')$ is the explanation vector for the perturbed version. For each applicant, this was calculated across all five perturbed versions and averaged into a single stability score

$$AvgStability = \frac{1}{m} \sum_{r=1}^m Stab(x, x'_r)$$

where m is the number of perturbed versions per applicant, five in this case, and x'_r is the r -th perturbed version. These per-applicant scores were then averaged again across all 50 applicants, giving one overall score per model, per explanation method, eight scores in total.

4.4.2 Similarity-Based Stability

This test looks at whether genuinely different applicants who happen to look alike receive similar explanations from the same trained model. 30 applicants were selected from the original 200 applicant sample to serve as anchors. For each anchor, the full validation set of 61,502 applicants was searched for its 5 nearest real neighbours, using the Euclidean distance calculated across all the 73 features after standardization. Standardization mattered here for the same reason it mattered for Logistic regression. Without it, a large magnitude feature like income would dominate the distance calculation, regardless of how relevant other features actually were to genuine similarity. Searching the full validation set, rather than just the original 200 applicant sample, was a deliberate choice to find the applicants who were genuinely similar, drawn from the widest possible pool of real people rather than whichever 200 happened to already be sampled. This search produced 150 unique neighbouring applicants, all outside the original 200 applicant sample, each needing a fresh SHAP and LIME explanation generated using the same above-described procedure. Two complementary metrics were then calculated for each anchor group, made of one anchor and its 5 neighbours.

The first is variance in feature importance, calculated per feature across the six explanation vectors in a group, then averaged across all features. This uses a standard sample variance formula:

$$Var_j = \frac{1}{n-1} \sum_{i=1}^n (\phi_{ij} - \bar{\phi}_j)^2$$

The second is rank consistency, calculated as the Spearman rank correlation (Spearman, 1904) between the anchor's feature importance ranking and each neighbour's ranking in turn, then averaged across the 5 neighbours.

$$\rho = 1 - \frac{6\sum d_i^2}{k(k^2 - 1)}$$

Here, d_i is the difference in rank for feature i between the two explanations being compared, and k is the number of features ranked. Feature rankings were based on the absolute size of each feature's contribution, consistent with how feature importance was treated throughout the explanation generation stage. Both metrics were averaged across all 30 anchor groups to produce one overall similarity stability score per model, per explanation method.

4.4.3 Cross-Model Comparison Framework

With stability scores calculated for all four models under both tests, the next step was to compare them properly and check whether any differences between models were real, rather than just the result of which applicants happened to be sampled.

The first step checks whether a real difference exists anywhere among the four models for a given metric, using a test called Kruskal Wallis. This test does not assume the scores are shaped like a normal bell curve, which suited the similarity and correlation scores used in this study.

Where a real difference was found, every model was then compared directly against every other model, one pair at a time. With four models, that gives six pairs in total, Logistic Regression against each of the other three, Decision Tree against Random Forest and XGBoost, and Random Forest against XGBoost. Each pair was tested using a second test called Mann-Whitney, to identify exactly which models differed from which.

Since six comparisons were being run at once, a standard adjustment called a Bonferroni correction was applied to each one, making every individual comparison stricter so that testing six pairs at once does not produce a false positive by chance.

The feature variance metric from the similarity test was left out of this comparison because Random Forest's contribution values were already smaller in scale than the other three models, so comparing raw variance across models would partly reflect that scale difference rather than genuine stability. Perturbation stability and rank consistency do not have this problem, since both measure similarity rather than raw size, and these two formed the basis of this comparison.

4.4.4 Performance-Stability Relationship

The final step was to relate each model's predictive performance to its explanation stability. ROC-AUC was used as the performance measure because it is the one metric in this study unaffected by Random Forest's different decision threshold, making it the fairest basis for comparing across models. Each model's ROC-AUC was paired with its perturbation stability and rank consistency scores from the two tests above, separately for SHAP and LIME. This gave eight performance stability pairs in total, four models under two explanation methods each.

With only four models, any correlation calculated between the performance and stability describes the pattern seen in this particular study rather than proving a general trade-off with strong statistical power. This is an accepted limitation of comparing model architectures rather than applicants. It cannot be solved the way the earlier tests were, by simply sampling more applicants, without moving away from the specific comparison of a linear model, a single tree, a bagged ensemble, and a boosted ensemble that this study was built around from the start.

Chapter 5: Results and Discussions

This chapter presents the findings obtained from our methodology. It is organized as follows. Section 5.1 presents the predictive performance of the four trained models, using accuracy, precision, recall, F1-score, and ROC-AUC. Section 5.2 examines how consistent each model's explanations remain when an applicant's own data is perturbed by a small amount. Section 5.3 examines whether genuinely similar applicants receive similar explanations from the same model. Section 5.4 brings these two stability tests together in a statistical comparison across all four models. Section 5.5 relates each model's predictive performance to its explanation stability, to establish whether a trade-off exists between the two. Section 5.6 illustrates these findings directly through individual explanation examples, including SHAP summary plots and a single applicant compared across all four models. Section 5.7 closes the chapter with a summary of the key findings and their limitations.

5.1 Predictive Model Performance

Table 5.1 shows how each of the four models performed on the validation set.

Table 5.1: Predictive Performance of the Four

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.6979	0.1667	0.6860	0.2683	0.7607
Decision Tree	0.6784	0.1482	0.6282	0.2398	0.6924
Random Forest	0.6942	0.1604	0.6584	0.2580	0.7369
XGBoost	0.7467	0.1819	0.6111	0.2803	0.7515

Random Forest is evaluated using a decision threshold of 0.0807, the real default rate in the training data, rather than the standard 0.5 used for the other three models. This adjustment was necessary because Random Forest's predicted probabilities were compressed toward the low end of the 0 to 1 range, a consequence of averaging predictions across 100 trees. ROC-AUC does not depend on this threshold and is unaffected by the difference.

Logistic regression achieves the highest ROC-AUC of the four models (as shown in Table 5.1), ahead of XGBoost, Random Forest, and Decision Tree in that order. This is a notable result on its own. Logistic Regression is the simplest and most transparent model in this study, yet it outperformed three more structurally complex models on the one metric that allows a fair comparison across all four. Decision Tree records the lowest ROC-AUC by a clear margin, consistent with the general expectation that a single, unaveraged tree generalises less reliably than an ensemble of many trees. The accuracy, precision, recall, and F1-score tell a related but separate story, shaped heavily by the class weighting applied during training. Recall is high across all four models, between 0.61 and 0.69, meaning each model correctly identifies a large share of applicants who actually default. Precision is low across all four, between 0.15 and 0.18, meaning most applicants flagged as high risk did not go on to default. This trade-off is the direct and expected result of weighting defaulter cases roughly 11.4 times more heavily during training, prioritising catching real defaulters over avoiding false alarms.

5.2 Perturbation Stability

A stable explanation should not change much when an applicant's own data is nudged only slightly. Each perturbed applicant was compared back against their own original explanation using cosine similarity, a score between 0 and 1 where 1 means the explanation stayed identical and 0 means it changed completely. Table 5.2 shows the average, lowest, and highest score recorded for each model and explanation method, across the 50 applicants tested.

Table 5.2: Perturbation Stability by Model and Explanation Method

Model	Method	Average	Lowest	Highest
Logistic Regression	SHAP	0.9959	0.9907	0.9982
Logistic Regression	LIME	0.9997	0.9997	0.9998

Decision Tree	SHAP	0.8522	0.4825	0.9861
Decision Tree	LIME	0.9382	0.9219	0.9522
Random Forest	SHAP	0.8502	0.6459	0.9674
Random Forest	LIME	0.9464	0.9360	0.9553
XGBoost	SHAP	0.9405	0.8194	0.9831
XGBoost	LIME	0.9686	0.9586	0.9761

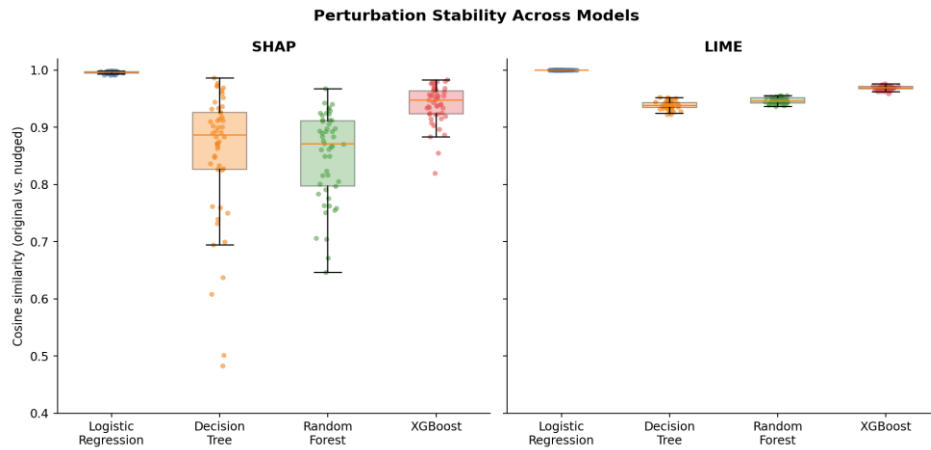


Figure 5.1: Perturbation stability by model, split by explanation method. Each point is one applicant’s average score across five perturbed versions.

Figure 5.1 above shows that Logistic regression is the most stable model under both explanation methods, with scores close to 1 for every applicant tested and almost no spread between the lowest and highest score recorded. LIME produced higher stability scores than SHAP for every single model, without exception. Decision tree shows the widest spread of any result in this table. Its SHAP scores range from 0.4825 up to 0.9861, meaning that for some applicants a small nudge barely changed the explanation, while for others it changed it almost completely. A plausible explanation is that Decision Tree makes its predictions using a sequence of hard yes or no splits, so a small change to an applicant’s data can occasionally push them across a threshold and down a completely different path through the tree, producing a very different explanation for what is still, in reality, almost the same applicant. LIME does not show this same volatility for Decision Tree, staying within a much narrower range of 0.9219 to 0.9522.

5.3 Similarity Stability

Genuinely similar applicants, real people who are not the same person but closely resemble one another, would be expected to receive similar explanations from the same model if that model’s explanations are trustworthy. Two measures were calculated for each of the 30 anchor applicants and their five nearest neighbours: feature importance variance and rank consistency. Table 5.3 shows both the feature variance between the five nearest neighbours and the rank consistency of these features averaged across all 30 anchor groups.

Table 5.3: Similarity Stability by Model and Explanation Method

Model	Method	Feature Variance	Rank Consistency
Logistic Regression	SHAP	0.000214	0.8932
Logistic Regression	LIME	0.000000	0.9954
Decision Tree	SHAP	0.000293	0.7821
Decision Tree	LIME	0.000011	0.3777
Random Forest	SHAP	0.000012	0.6497
Random Forest	LIME	0.000000	0.6279
XGBoost	SHAP	0.000229	0.7549
XGBoost	LIME	0.000006	0.7870

The feature variance column is not directly comparable across models. Random Forest’s raw contribution values are naturally smaller than the other three models, a consequence of averaging predictions across 100 trees, so its

lower variance partly reflects this difference in scale rather than genuinely steadier explanations. Rank consistency does not have this problem, since it measures agreement in ordering rather than the raw size of a contribution, and is the more reliable of the two measures for comparing across models.

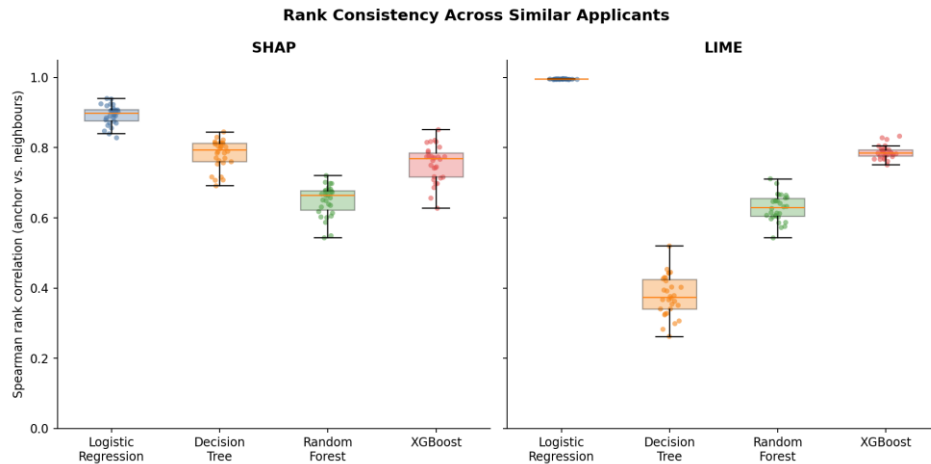


Figure 5.2: Rank consistency by model, split by explanation method. Each point is one anchor applicant's average agreement with its five nearest neighbours.

As shown in Figure 5.2, Logistic regression again produces the most consistent explanations of the four models, under both methods. The clearest result in this table, and one of the clearest in the study, is Decision Tree under LIME, which drops to 0.3777, by far the lowest score recorded anywhere in this section. This held consistently across all 30 anchor groups tested, ranging narrowly between 0.26 and 0.52, meaning that for this specific combination of model and explanation method, two applicants who look alike on paper were, more often than not, given genuinely different accounts of what mattered in their assessment. This result stands in contrast to Section 5.2, where Decision Tree under LIME was among the more stable combinations tested. The same model and the same explanation method behaved very differently depending on which kind of stability was being tested, small nudges to one applicant versus comparisons between different applicants who resemble each other. A plausible reason is that LIME builds its explanation by sampling many synthetic points close to the applicant being explained. When the applicant themselves barely changes, that local neighbourhood barely changes either. When two different applicants are being compared, LIME is sampling around two different points in the data, and any sharp threshold in the Decision Tree sitting between them can produce two locally accurate but genuinely different explanations.

5.4 Comparing Stability Across Models

The above results suggest some real differences in stability between the four models. Whether those differences are statistically genuine was checked using an overall test, run first for each metric and each explanation method, to check whether a real difference existed anywhere among the four models. All four overall tests returned a result below the standard 0.05 threshold for statistical significance, confirming that genuine differences exist in every case. Table 5.4 summarises the pairwise comparisons that followed.

Table 5.4: Statistical Comparison of Stability Across Models

Comparison	Perturbation (SHAP)	Perturbation (LIME)	Rank Consistency (SHAP)	Rank Consistency (LIME)
Logistic Regression vs Decision Tree	Significant	Significant	Significant	Significant
Logistic Regression vs Random Forest	Significant	Significant	Significant	Significant
Logistic Regression vs XGBoost	Significant	Significant	Significant	Significant
Decision Tree vs Random Forest	Not significant	Significant	Significant	Significant

Decision Tree vs XGBoost	Significant	Significant	Not significant	Significant
Random Forest vs XGBoost	Significant	Significant	Significant	Significant

Logistic regression is significantly more stable than every other model, on every metric, under both explanation methods, without a single exception across all eight comparisons involving it. This is the most consistent finding in the study. Two pairs of models could not be statistically distinguished from one another. Decision Tree and Random Forest showed no significant difference in SHAP perturbation stability, and Decision Tree and XGBoost showed no significant difference in SHAP rank consistency. Every other comparison across the table was significant. XGBoost is worth highlighting specifically. As the most structurally complex of the four models, it might be expected to produce the least stable explanations. It did not. XGBoost was significantly more stable than both Decision Tree and Random Forest on perturbation stability under both explanation methods, and was statistically indistinguishable from Decision Tree, rather than clearly worse, on SHAP rank consistency. The model complexity alone does not appear to predict the explanation instability in these results.

5.5 Predictive Performance and Explanation Stability

Table 5.5 restates each model's ROC-AUC from Section 5.1 alongside its average perturbation stability and rank consistency scores from Sections 5.3 and 5.4, bringing predictive performance and explanation stability together for direct comparison.

Table 5.5: Predictive Performance and Explanation Stability Combined

Model	Method	ROC-AUC	Perturbation Stability	Rank Consistency
Logistic Regression	SHAP	0.7607	0.9959	0.8932
Logistic Regression	LIME	0.7607	0.9997	0.9954
Decision Tree	SHAP	0.6924	0.8522	0.7821
Decision Tree	LIME	0.6924	0.9382	0.3777
Random Forest	SHAP	0.7369	0.8502	0.6497
Random Forest	LIME	0.7369	0.9464	0.6279
XGBoost	SHAP	0.7515	0.9405	0.7549
XGBoost	LIME	0.7515	0.9686	0.7870

Figures 5.3 and 5.4 plot this relationship directly, showing predictive performance against perturbation stability and against rank consistency respectively, one point per model per explanation method.

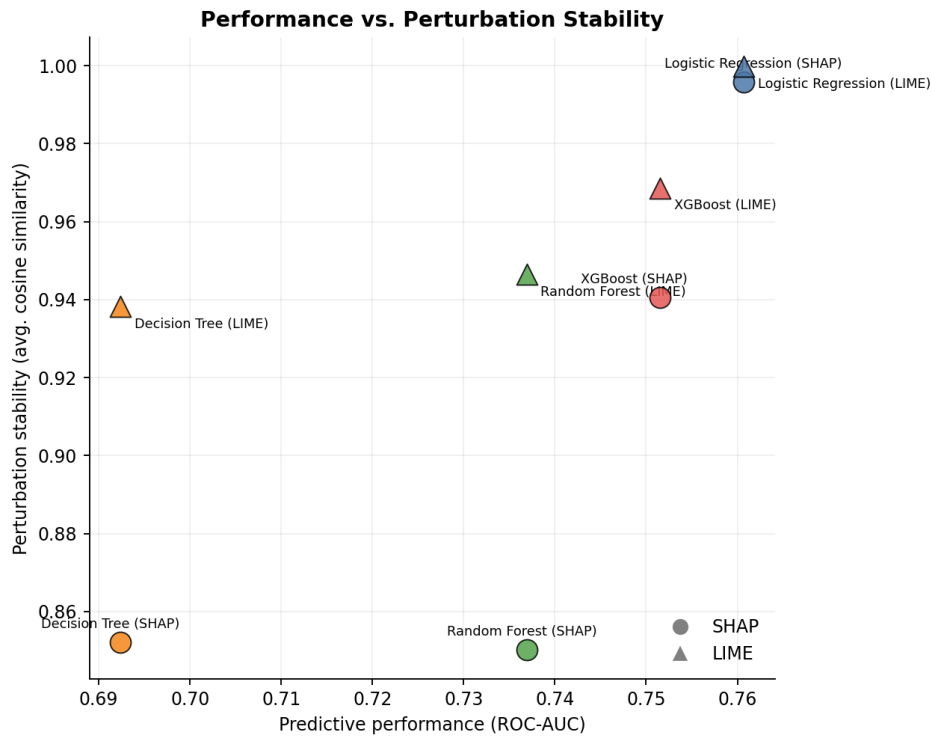


Figure 5.3: Predictive performance against perturbation stability, one point per model per explanation method

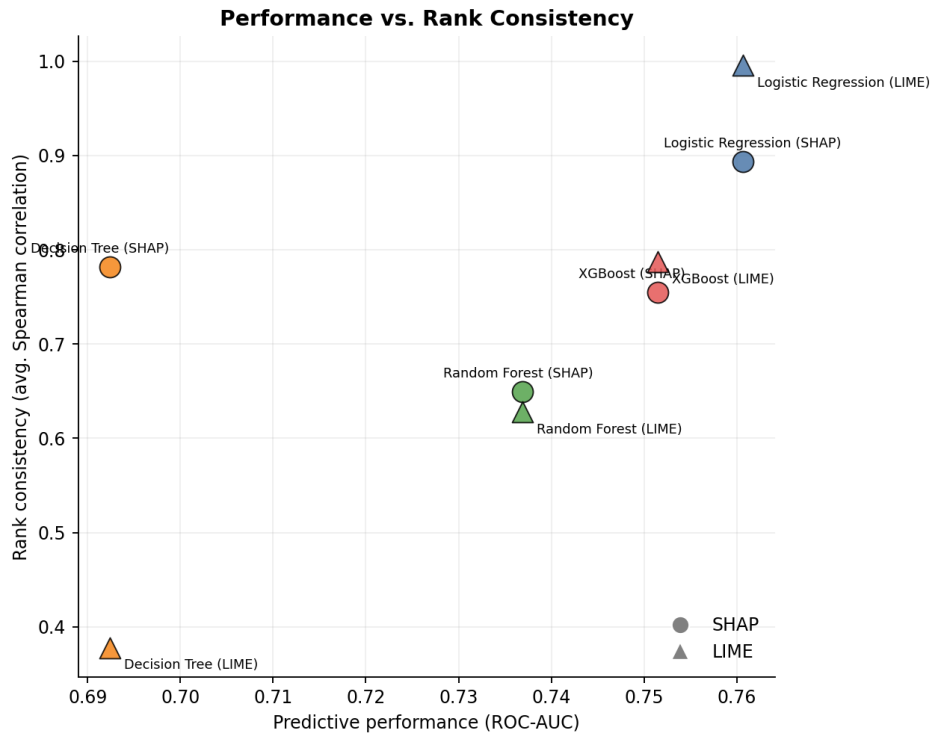


Figure 5.4: Predictive performance against rank consistency, one point per model per explanation method.

The Pearson correlation was used to summarize the relationship shown in Figures 5.3 and 5.4 between the performance and stability. The performance against the perturbation stability produced a correlation of 0.630, and the performance against the rank consistency produced a correlation of 0.676. Both are positive, meaning that the studied models with higher ROC-AUC tended to have more stable explanations, an opposite direction from the trade-off commonly assumed in much of the literature reviewed earlier. Neither correlation reached the standard threshold for statistical significance, with p-values of 0.094 and 0.066 respectively.

This result should be read carefully. With only four underlying models, a correlation calculated across eight points describes a pattern rather than proving a general relationship with strong statistical confidence. Much of the positive pattern is driven by Logistic regression, which combines the highest predictive performance with the highest stability of any tested model. Rather than a general rule that more accurate models are always more stable, the more defensible reading of this result is that this study found no evidence of the assumed trade-off, and the simplest, most interpretable model performed best on both measures at once.

5.6 Individual Explanation Examples

The results so far describe stability in aggregate, across many applicants and many comparisons. The same patterns are shown directly here, first through SHAP summary plots covering all 200 sampled applicants for each model, then through a single applicant's LIME explanation compared across all four models.

5.6.1 SHAP Summary Plots

A SHAP summary plot places every sampled applicant as a single dot for each feature, positioned by how much that feature pushed their predicted risk up or down, and coloured by whether that applicant's actual value for that feature was high or low. Figures 5.5 5.8 show this for each of the four models.

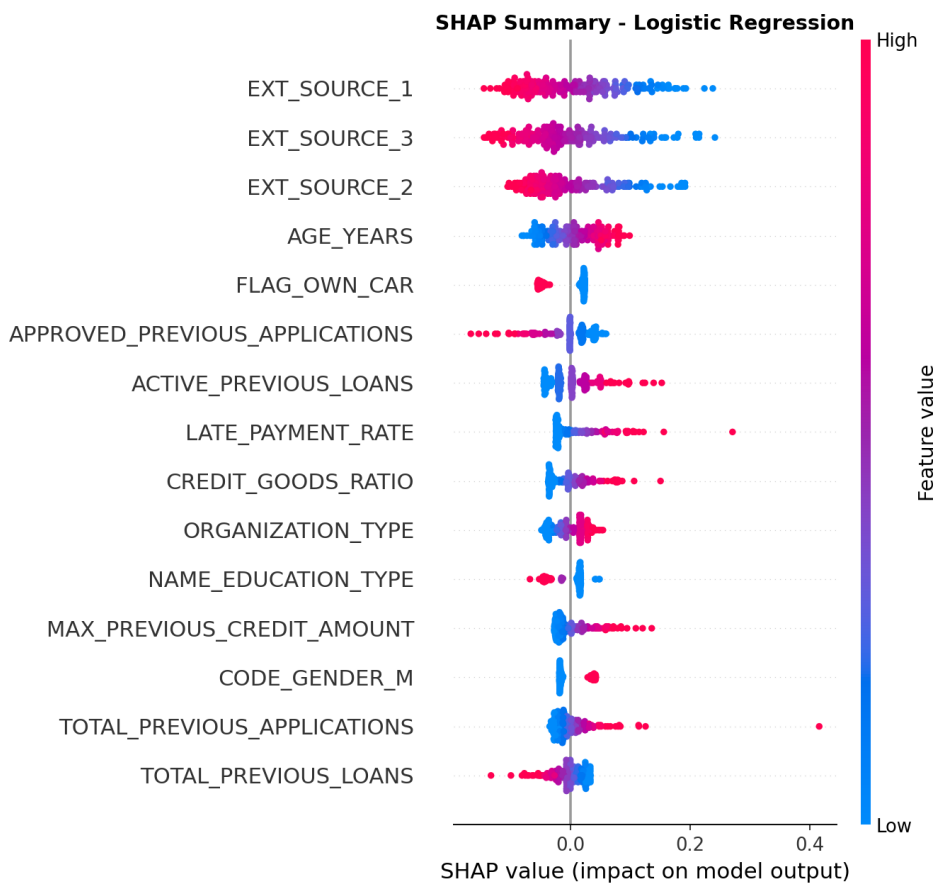


Figure 5.5: SHAP summary plot, Logistic Regression.

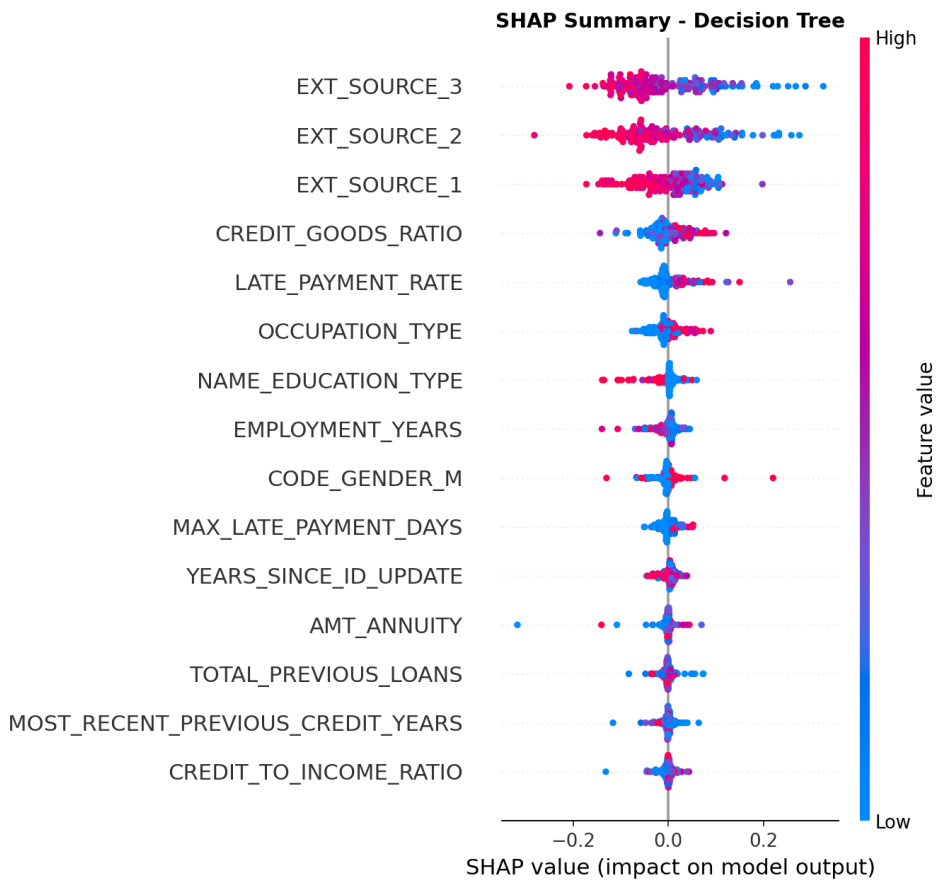


Figure 5.6: SHAP summary plot, Decision Tree.

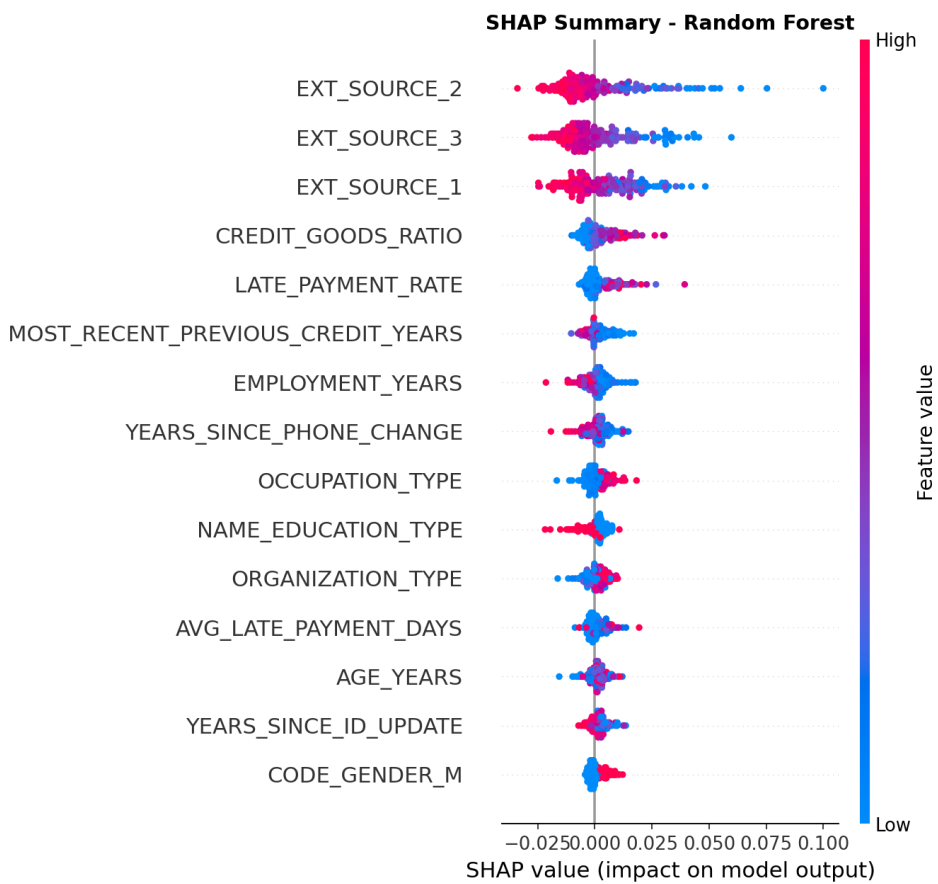


Figure 5.7: SHAP summary plot, Random Forest.

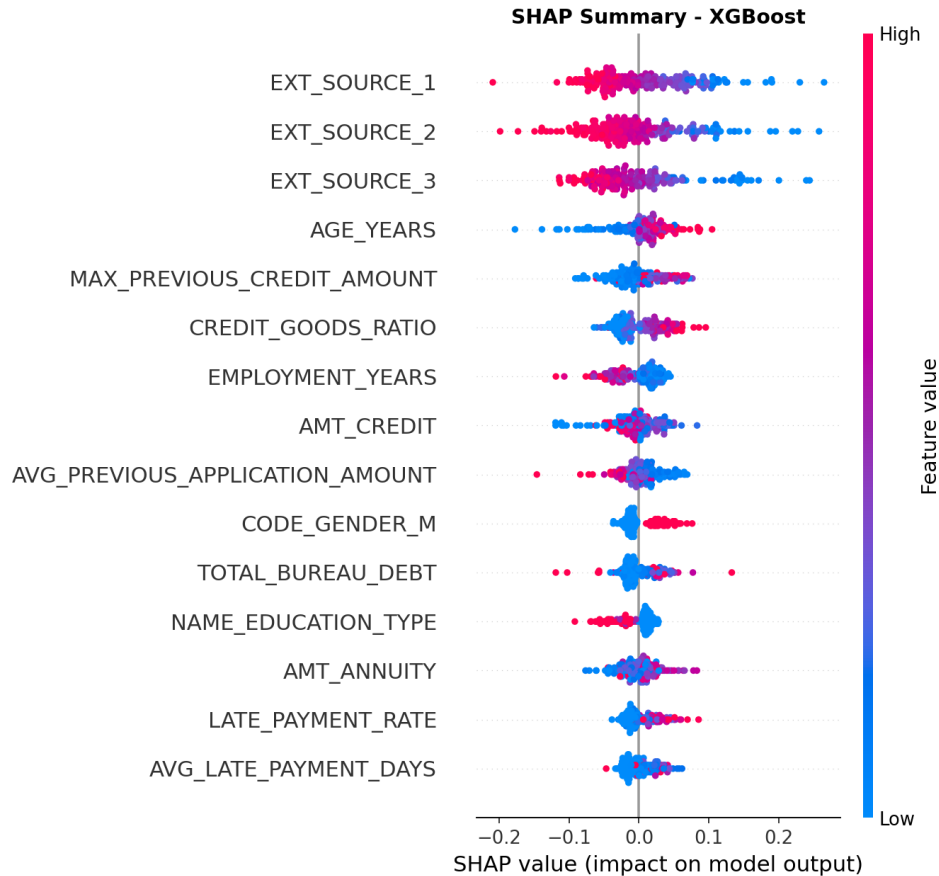


Figure 5.8: SHAP summary plot, XGBoost.

EXT_SOURCE_1, EXT_SOURCE_3, and EXT_SOURCE_2 dominate Logistic Regression's plot in Figure 5.5, each showing a consistent direction: a high score, shown in red, reduces predicted risk, while a low score, shown in blue, increases it.

The Decision Tree's plot in Figure 5.6 shows individual points for EXT_SOURCE_3 extending unusually far to the right, beyond 0.3, further than any single point for that feature in any other model. This lines up with the wide spread already found for Decision Tree under SHAP in Section 5.2, the highest standard deviation recorded anywhere in the perturbation test, and gives a visual sense of why that spread exists: a small number of applicants receive an unusually large contribution from this one feature.

The compression in the random Forest's predicted probabilities (discussed in Section 5.2) is directly visible in Figure 5.7. Its horizontal axis spans only roughly -0.025 to 0.100, compared to a range of roughly -0.2 to 0.3 or wider for the other three models. The same underlying effect that required a recalibrated decision threshold for Random Forest is visible here as a genuinely narrower range of feature contributions.

XGBoost's plot shown in Figure 5.9 show that EXT_SOURCE_1, EXT_SOURCE_2, and EXT_SOURCE_3 remain the top three features, though AGE_YEARS and MAX_PREVIOUS_CREDIT_AMOUNT also register a meaningful contribution, features that did not appear as prominently in the other three models.

In summary, EXT_SOURCE_1, EXT_SOURCE_2, and EXT_SOURCE_3 sit at the top of all four plots, and the colouring is consistent across every model: a low score, shown in blue, sits on the risk increasing side, while a high score, shown in red, sits on the risk reducing side. This matches the negative correlation between these three features and TARGET already established in the exploratory data analysis, now confirmed directly in how each trained model actually uses them.

5.6.2 Comparing One Applicant Across All The Four Models

Figure 5.9 shows the LIME's explanation for the same applicant (applicant 377489) generated separately by all the four models.

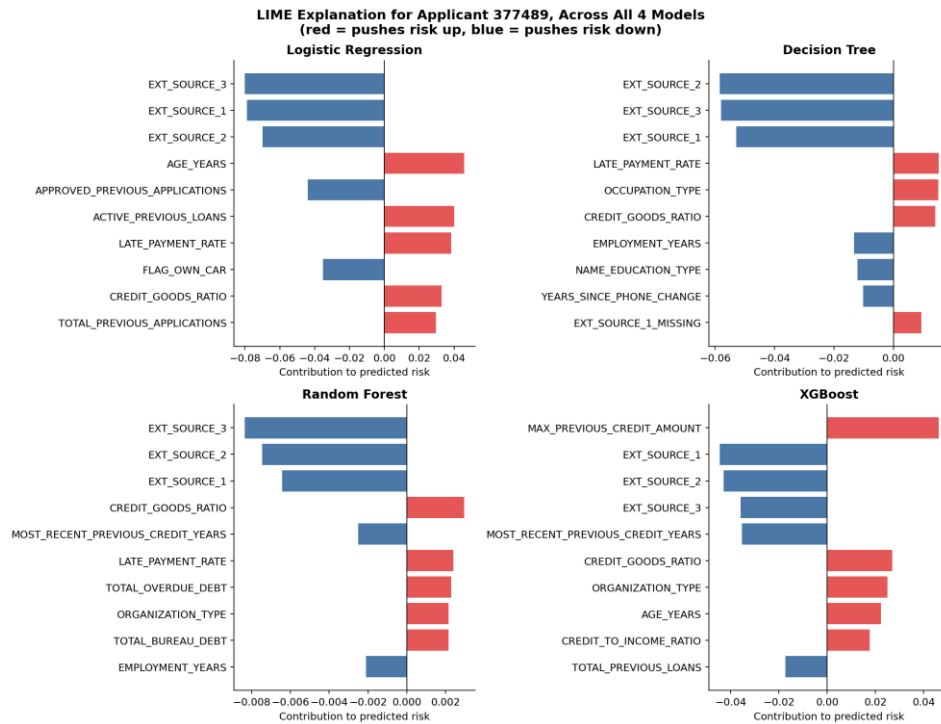


Figure 5.9: LIME explanation for a single applicant, across all four models. – Ensure that this figure is **cited** in the text

As shown in Figure 5.9, the Logistic regression, Decision Tree, and Random Forest agree closely for this applicant. All three rank the three EXT_SOURCE scores as the top factors reducing their risk, in the same order in two of the three models. XGBoost breaks this pattern. Its top ranked feature for this same applicant is MAX_PREVIOUS_CREDIT_AMOUNT, a feature that does not appear at all among the top factors for the other three models, and it ranks all three EXT_SOURCE scores below it. This is a concrete, applicant level example of a pattern already visible in the aggregate results: XGBoost's SHAP and LIME rankings were less consistent with one another than the other three models across this study, and here that disagreement is visible in the explanation of one real person rather than only as a statistic.

In the Logistic Regression explanation for this applicant, AGE_YEARS pushes predicted risk up, while the exploratory data analysis found that, across the dataset as a whole, older applicants default less often on average. This is not necessarily a contradiction. A model's contribution for one feature, for one applicant, reflects that feature's effect after accounting for every other feature about that same applicant at once, which is a different calculation from a simple, dataset wide relationship between age and default. AGE_YEARS and EMPLOYMENT_YEARS are correlated at 0.76 in this dataset, a strong relationship, and when two features move together this closely, a linear model can struggle to cleanly divide credit between them, which can shift an individual applicant's result away from the broader population trend even while that trend holds overall. Without this applicant's full underlying data, this is offered as a plausible explanation for the pattern rather than a confirmed one.

5.7 Summary

Logistic Regression produced the most stable explanations of the four models tested, under both SHAP and LIME, across every metric and every statistical comparison in this study, while also achieving the highest ROC-AUC. LIME produced more stable explanations than SHAP for every model on the perturbation test, a consistent pattern across the entire study. Decision Tree combined with LIME produced a striking result: weak and consistent agreement between similar applicants despite being one of the more stable combinations under small perturbations, a reversal that points to LIME and SHAP responding differently to different kinds of instability rather than one method simply being better than the other. XGBoost, despite being the most

structurally complex model in this study, was not the least stable, and in several comparisons could not be statistically distinguished from far simpler models, a pattern also visible directly in its individual explanation output. No evidence was found of a general trade-off between predictive performance and explanation stability, and the relationship observed pointed in the opposite direction, though this should be read as a pattern specific to this study rather than a proven general rule.

5.7.1 Limitations

The performance and stability relationship examined in Section 5.5 is based on only four models, which limits how confidently a general trade-off can be confirmed or ruled out. The perturbation and similarity tests were each run on a sample of applicants rather than the full validation set, for computational reasons, and while the samples were chosen to be representative, a larger sample would allow finer detail in results such as the spread seen for Decision Tree under SHAP. Feature importance variance, one of the two similarity measures used in this study, could not be compared directly across models due to differences in the raw scale of each model's contribution values, and rank consistency was relied upon instead for cross-model comparisons involving that test. The individual explanation examples in Section 5.6 are illustrative of patterns already established statistically elsewhere in this chapter, and are drawn from a single applicant and a single sample of 200 applicants, rather than intended as findings in their own right.

Chapter 6: Conclusion and Future Work

6.1 Conclusion

This study was conducted to improve the trust in credit scoring systems, by evaluating the stability and consistency of explainable AI methods. Four credit scoring models spanning a real range of structural complexity, were explored, namely: Logistic regression, a single Decision tree, Random forest, and XGBoost, were trained on a dedicated dataset, explained using both SHAP and LIME, and tested for how stable those explanations were under small input changes and across genuinely similar applicants. Each research question set out at the start of this study can now be answered directly as follows.

- (a) How consistent are SHAP and LIME explanations for the same applicant profile under small input perturbations?** Consistency varied substantially by model. The logistic regression’s explanations were almost unaffected by a small nudge to an applicant’s own data, under both SHAP and LIME. The decision tree’s SHAP explanations were the least consistent of any combination tested, with some applicants’ explanations barely changing and others changing almost completely from the same size of nudge. LIME was more consistent than SHAP for every model under this test, without exception.
- (b) Do similar applicants receive similar explanations under the same trained credit-scoring model?** Generally, yes, with one clear exception. Logistic regression again produced the most consistent explanations across genuinely similar applicants. The decision tree combined with LIME broke this pattern sharply, producing weak agreement between similar applicants despite having been comparatively stable under the perturbation test, a reversal that shows explanation stability is not a single property a model either has or lacks, but something that can depend on what kind of change is being tested.
- (c) How does explanation stability differ between Logistic Regression and black-box models such as XGBoost?** Logistic regression was significantly more stable than every other model, on every metric tested, under both explanation methods, with no exceptions across any of the statistical comparisons carried out. XGBoost, despite being structurally the most complex model in this study, was not the least stable of the black-box models, and on some measures could not be statistically distinguished from a single decision tree. Structural complexity alone did not reliably predict how stable a model’s explanations would be.
- (d) Does higher predictive performance correspond to more stable explanations, or is there a trade-off between predictive accuracy and explanation stability?** This study found no evidence of the trade-off often assumed in the literature. The relationship observed between predictive performance and explanation stability was positive rather than negative, largely driven by the logistic regression, which achieved both the highest ROC-AUC and the highest stability of any model tested. With only four models compared, this result describes a pattern specific to this study rather than a statistically confirmed general rule, but it is a meaningful counterpoint to the assumption that explainability must be traded off against performance.

Taken together, these findings support the original aim of this study. Explanation stability is not a property that can be assumed once the SHAP or LIME has been applied; it varies meaningfully by model, by explanation method, and by the kind of change being tested against. For a domain like credit scoring, where an explanation may need to be defended to a regulator or a declined applicant, this variation matters as much as a model’s raw predictive accuracy. The simplest, most transparent model in this study was also the most trustworthy by every stability measure used, a result worth real weight in any decision about which model to deploy in a setting where explanations themselves need to be relied upon.

6.2 Contribution of This Study

Much of the existing literature on explainable AI in credit scoring treats SHAP and LIME as reliable once applied, without directly testing that assumption on structured, tabular financial data. This study brought together three strands of literature that have rarely been combined in a single and systematic study of credit scoring specifically: the applied use of SHAP and LIME in credit risk, the general machine learning literature documenting explanation instability, and the comparison of interpretable and black-box model architectures. In

doing so, this study has produced a concrete, statistically grounded answer to a question that the existing literature had largely assumed rather than tested, using a perturbation-based test and a real-world similarity test applied consistently across four dedicated models and two explanation methods.

6.3 Limitations

Beyond the specific limitations already noted in the discussion of individual results, several broader limitations apply to this study as a whole. All findings have been based on a single dataset, the Home Credit Default Risk dataset; the explanation stability could behave differently on a dataset with a different feature set, class balance, or lending context, and this study cannot speak to whether these patterns generalize beyond it.

The perturbation test used a single nudge size of 5%, chosen as a reasonable, realistic definition of a small change. A different nudge size could produce a different picture of stability, particularly for the decision tree model, where the instability observed appeared to be driven by the applicants sitting close to a specific decision threshold; a smaller or larger nudge could plausibly shift how many applicants fall into that category.

This study has also evaluated whether the explanations were stable, but did not test whether the differences found on these explanations would actually be noticed or considered meaningful, by the credit analysts, regulators, or applicants who would encounter these explanations in practice. A difference that is statistically significant is not automatically a difference that matters to the person reading the explanation.

6.4 Future Works

Several directions of future works pertaining to the scope of this report include:

- Testing across multiple credit datasets with different feature sets and class balances, to establish whether the stability patterns found here, particularly the logistic regression’s consistent advantage and the decision tree’s instability under SHAP, can generalize beyond this one dataset.
- Repeating the carried perturbation test at multiple nudge sizes, in order to establish whether instability in tree-based models scales predictably with the size of the input change, or whether it is dominated by threshold effects at a particular scale.
- Extending the considered stability framework to additional model families such as gradient boosting variants beyond XGBoost or simple neural networks, to test whether the lack of a performance-stability trade-off found here holds more broadly.
- Investigating the targeted fixes for the specific instability found in the decision tree model, such as the stabilised variants of LIME proposed in the wider interpretability literature, and testing whether they resolve the sharp drop in rank consistency found here without sacrificing the model’s inherent transparency.
- Conducting a human-subject study with credit analysts or loan applicants, to establish what degree of explanation variation is actually noticed or considered material, complementing the statistical stability measures used in this study with a measure of its practical significance.

Appendix A: Data Dictionary

This appendix lists every feature present in the final modelling dataset (train_selected.csv / test_selected.csv), organised by source table, with a description and derivation for each.

A.1 Application-Level Features

Covers every applicant-level column present in the final merged dataset. TARGET is present only in the training data. Table A.1 shows the Application-Level features with their description and how they were derived.

Table A.1: Application-Level Features

Feature	Description	Derivation
SK_ID_CURR	Unique applicant identifier.	
TARGET	1 = applicant had payment difficulty/default. 0 = no payment difficulty. Present only in train_merged.csv.	
NAME_CONTRACT_TYPE	Type of loan contract requested.	0 = Cash loans, 1 = Revolving loans.
FLAG_OWN_CAR	Whether the applicant owns a car.	0 = No, 1 = Yes.
FLAG_OWN_REALTY	Whether the applicant owns a house or flat.	0 = No, 1 = Yes.
CNT_CHILDREN	Number of children the applicant has.	
AMT_INCOME_TOTAL	Total income of the applicant.	
AMT_CREDIT	Total credit amount requested or granted for the current loan.	
AMT_ANNUITY	Regular repayment amount for the current loan.	12 originally-missing rows filled with the training-set median.
NAME_TYPE_SUITE	Who accompanied the applicant when applying.	Smoothed target encoding: category replaced with its average TARGET rate, weighted toward the overall training rate by category size. Missing values grouped as “Unknown” before encoding.
NAME_INCOME_TYPE	Applicant’s income type (e.g. Working, Pensioner, Commercial associate).	Smoothed target encoding.
NAME_EDUCATION_TYPE	Highest education level achieved.	Ordinal: 0 = Lower secondary, 1 = Secondary / secondary special, 2 = Incomplete higher, 3 = Higher education, 4 = Academic degree.
NAME_FAMILY_STATUS	Family or marital status of the applicant.	Smoothed target encoding.
NAME_HOUSING_TYPE	Applicant’s housing situation.	Smoothed target encoding.
REGION_POPULATION_RELATIVE	Normalized population level of the applicant’s region. Higher = more populated.	
OWN_CAR_AGE	Age of the applicant’s car, if owned.	Left blank (NaN) for applicants with no car (FLAG_OWN_CAR = 0) and for a small number of car owners with a genuinely unrecorded age.
FLAG_EMP_PHONE	Whether the applicant provided an employment phone number.	
FLAG_PHONE	Whether the applicant provided a phone number.	

FLAG_EMAIL	Whether the applicant provided an email address.	
OCCUPATION_TYPE	Applicant's occupation category.	Smoothed target encoding. Originally-missing values ("Unknown") encoded as their own category.
CNT_FAM_MEMBERS	Number of family members in the applicant's household.	2 originally-missing rows filled with the training-set median.
REGION_RATING_CLIENT_W_CITY	Regional risk rating adjusted with city-level information (1 = Low Risk, 3 = High Risk).	
HOUR_APPR_PROCESS_START	Hour of the day the applicant submitted the loan application.	
REG_REGION_NOT_WORK_REGION	Whether the applicant's registered region differs from their work region.	
LIVE_REGION_NOT_WORK_REGION	Whether the applicant's living region differs from their work region.	
REG_CITY_NOT_WORK_CITY	Whether the applicant's registered city differs from their work city.	
LIVE_CITY_NOT_WORK_CITY	Whether the applicant's living city differs from their work city.	
ORGANIZATION_TYPE	Type of organization where the applicant works.	Smoothed target encoding.
EXT_SOURCE_1	Anonymized normalized score from an external data source.	Genuinely-missing values (56.4%) imputed with a linear regression model trained on applicant age, income, credit amount, employment years, EXT_SOURCE_2, and EXT_SOURCE_3, fitted on training data only.
EXT_SOURCE_1_MISSING	Whether EXT_SOURCE_1 was originally missing before imputation.	1 if missing prior to imputation, else 0.
EXT_SOURCE_2	Anonymized normalized score from an external data source.	Genuinely-missing values (0.2%) filled with the training-set median.
EXT_SOURCE_2_MISSING	Whether EXT_SOURCE_2 was originally missing before imputation.	1 if missing prior to imputation, else 0.
EXT_SOURCE_3	Anonymized normalized score from an external data source.	Genuinely-missing values (19.8%) imputed with a linear regression model trained on applicant age, income, credit amount, employment years, and EXT_SOURCE_2, fitted on training data only.
EXT_SOURCE_3_MISSING	Whether EXT_SOURCE_3 was originally missing before imputation.	1 if missing prior to imputation, else 0.
OBS_30_CNT_SOCIAL_CIRCLE	Number of observations in the applicant's social circle with 30 days past due.	Originally-missing rows (0.33%) filled with the training-set median.
DEF_30_CNT_SOCIAL_CIRCLE	Number of observations in the applicant's social circle that defaulted at 30 days past due.	Originally-missing rows filled with the training-set median.

DEF_60_CNT_SOCIAL_CIRCLE	Number of observations in the applicant's social circle that defaulted at 60 days past due.	Originally-missing rows filled with the training-set median.
SOCIAL_CIRCLE_MISSING	Whether the applicant's social circle counts were originally missing.	1 if all four social circle columns were missing, else 0.
AMT_REQ_CREDIT_BUREAU_HOUR	Number of credit bureau enquiries about the applicant in the hour before application.	Originally-missing rows (13.5%, applicants with no bureau history) filled with 0.
AMT_REQ_CREDIT_BUREAU_DAY	Number of credit bureau enquiries in the day before application.	Originally-missing rows filled with 0.
AMT_REQ_CREDIT_BUREAU_WEEK	Number of credit bureau enquiries in the week before application.	Originally-missing rows filled with 0.
AMT_REQ_CREDIT_BUREAU_MON	Number of credit bureau enquiries in the month before application.	Originally-missing rows filled with 0.
AMT_REQ_CREDIT_BUREAU_QRT	Number of credit bureau enquiries in the quarter before application.	Originally-missing rows filled with 0.
AMT_REQ_CREDIT_BUREAU_YEAR	Number of credit bureau enquiries in the year before application.	Originally-missing rows filled with 0.
AGE_YEARS	Applicant's age in years.	$AGE_YEARS = (-DAYS_BIRTH) / 365$.
EMPLOYMENT_YEARS	Number of years since the applicant started their current employment.	$EMPLOYMENT_YEARS = (-DAYS_EMPLOYED) / 365$, set to 0 for Pensioner/Unemployed applicants (NAME_INCOME_TYPE), for whom a current employment duration does not apply.
REGISTRATION_YEARS	Number of years since the applicant last changed registration.	$REGISTRATION_YEARS = (-DAYS_REGISTRATION) / 365$.
YEARS_SINCE_ID_UPDATE	Number of years since the applicant last changed their identity document.	$YEARS_SINCE_ID_UPDATE = (-DAYS_ID_PUBLISH) / 365$.
YEARS_SINCE_PHONE_CHANGE	Number of years since the applicant last changed phone number.	$YEARS_SINCE_PHONE_CHANGE = (-DAYS_LAST_PHONE_CHANGE) / 365$. 1 originally-missing row filled with the training-set median.
CREDIT_TO_INCOME_RATIO	Loan amount relative to applicant income.	$AMT_CREDIT / AMT_INCOME_TOTAL$.
ANNUITY_TO_INCOME_RATIO	Regular repayment burden relative to applicant income.	$AMT_ANNUITY / AMT_INCOME_TOTAL$. 12 originally-missing rows (inherited from AMT_ANNUITY) filled with the training-set median.
CREDIT_GOODS_RATIO	Loan amount relative to the price of the goods being financed. Replaces AMT_GOODS_PRICE (0.987 correlated with AMT_CREDIT).	$AMT_CREDIT / AMT_GOODS_PRICE$. Missing ratios filled with the training-set median ratio.
CODE_GENDER_M	Whether the applicant's recorded gender is Male.	One-hot indicator, fit on training categories.

A.2 Bureau History (Aggregated)

Table A.2 summarises each applicant's external credit bureau history from bureau.csv, aggregated to one row per applicant.

Table A.2: Bureau History (Aggregated) Features

Feature	Description	Derivation
TOTAL_PREVIOUS_LOANS	Total number of previous bureau credit records for the applicant.	Count of SK_ID_BUREAU per SK_ID_CURR.
ACTIVE_PREVIOUS_LOANS	Number of previous bureau credits still active.	Count of CREDIT_ACTIVE = "Active" per SK_ID_CURR.
TOTAL_BUREAU_DEBT	Total outstanding debt across the applicant's previous bureau credits.	Sum of AMT_CREDIT_SUM_DEBT per SK_ID_CURR.
TOTAL_OVERDUE_DEBT	Total overdue debt across the applicant's previous bureau credits.	Sum of AMT_CREDIT_SUM_OVERDUE per SK_ID_CURR.
MAX_DAYS_OVERDUE	Worst recorded overdue delay in previous bureau credits.	Max of CREDIT_DAY_OVERDUE per SK_ID_CURR.
MOST_RECENT_PREVIOUS_CREDIT_YEARS	How recently the applicant took a previous bureau credit.	Minimum of (-DAYS_CREDIT / 365) per SK_ID_CURR.
COUNT_CREDIT_CARDS	Number of previous credit card records.	Count of CREDIT_TYPE = "Credit card" per SK_ID_CURR.

A.3 Previous Home Credit Applications (Aggregated)

Table A.3 summarises the applicant's past Home Credit application history from previous_application.csv, aggregated to one row per applicant.

Table A.3: Previous Home Credit Application (Aggregated) Features

Feature	Description	Derivation
TOTAL_PREVIOUS_APPLICATIONS	Total number of previous Home Credit applications made by the applicant.	Count of SK_ID_PREV per SK_ID_CURR.
APPROVED_PREVIOUS_APPLICATIONS	Number of previous applications that were approved.	Count of NAME_CONTRACT_STATUS = "Approved" per SK_ID_CURR.
REFUSED_PREVIOUS_APPLICATIONS	Number of previous applications that were refused.	Count of NAME_CONTRACT_STATUS = "Refused" per SK_ID_CURR.
PREVIOUS_APPROVAL_RATE	Proportion of previous applications that were approved.	$\frac{\text{APPROVED_PREVIOUS_APPLICATIONS}}{\text{TOTAL_PREVIOUS_APPLICATIONS}}$

PREVIOUS_REFUSAL_RATE	Proportion of previous applications that were refused.	REFUSED_PREVIOUS_APPLICATIONS / TOTAL_PREVIOUS_APPLICATIONS .
AVG_PREVIOUS_APPLICATION_AMOUNT	Average amount the applicant previously applied for.	Mean of AMT_APPLICATION per SK_ID_CURR.
MAX_PREVIOUS_CREDIT_AMOUNT	Largest previous credit amount granted to the applicant.	Max of AMT_CREDIT per SK_ID_CURR.

A.4 Instalment Repayment History (Aggregated)

Table A.4 summarises actual repayment behaviour from installments_payments.csv, aggregated to one row per applicant.

Table A.4: Instalment Repayment History (Aggregated) Features

Feature	Description	Derivation
TOTAL_INSTALLMENT_RECORDS	Total number of installment/payment records available for the applicant.	Count of SK_ID_PREV per SK_ID_CURR.
LATE_PAYMENT_COUNT	Number of installment payments made after the scheduled due date.	Count of records where (DAYS_ENTRY_PAYMENT - DAYS_INSTALLMENT) > 0.
LATE_PAYMENT_RATE	Proportion of the applicant's installment payments that were late.	LATE_PAYMENT_COUNT / TOTAL_INSTALLMENT_RECORDS.
AVG_LATE_PAYMENT_DAYS	Average late-payment delay in days. Early/on-time payments count as 0.	Mean of max(DAYS_ENTRY_PAYMENT - DAYS_INSTALLMENT, 0) per SK_ID_CURR.
MAX_LATE_PAYMENT_DAYS	Worst/longest late-payment delay recorded. Early/on-time payments count as 0.	Max of max(DAYS_ENTRY_PAYMENT - DAYS_INSTALLMENT, 0) per SK_ID_CURR.

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